

Chain of Thought Reduces Errors from Misleading Names in Arithmetic

Anonymous ACL submission

Abstract

Renaming a variable should not change an arithmetic answer. Yet a model may answer 4 because a variable is named four, even when its equations give 6. We ask whether writing the calculation before answering reduces such errors. We extend Kudo et al.’s study of how language models compute intermediate values during chain-of-thought reasoning. Our problems add misleading variable names while keeping the equations and correct answers fixed. Across thirteen models and three sets of worked examples, we compare direct answers with reasoning chains that restate and evaluate each equation. Without written steps, renaming changes accuracy or the frequency of name-suggested answers in eleven models. With steps, the increase in name-suggested answers over ordinary names falls within ± 2 percentage points in 43 of 44 comparisons. Classifiers trained to recover values from model states do not establish why: their association with errors depends on one low-accuracy model, and they perform poorly on its generated steps even after training on such steps. The experiments support a behavioral reduction while leaving its internal cause unresolved.

1 Introduction

In the problem

pen=1 + cup, cup=2 + 3; pen=?

pen has value 6. Renaming it four leaves the answer unchanged:

four=1 + cup, cup=2 + 3; four=?

The definition says 6; the name suggests 4. We call 4 the *name-suggested value* and answering 4 a *name error*. That digit occurs nowhere in the equations and differs from every variable’s value, so the error points to the name rather than a copied digit.

Misleading identifiers and comments reduce accuracy on code tasks (Le et al., 2025; Lam et al., 2025; Le et al., 2026). Accuracy alone cannot locate the error: the model may misread the input, compute the wrong value, or answer incorrectly despite retaining the right value. Named-variable arithmetic lets us compare the computed and name-suggested values. Kudo et al. (2026) use probes and patching to study when values become available during chain-of-thought (CoT) arithmetic. We add names that disagree with those values. This cue-conflict design, used in Stroop tasks and shape–texture studies (Stroop, 1935; Geirhos et al., 2019), gives the name and the computation distinct answers.

Writing a chain usually reduces name errors. Without CoT, names affect answers in eleven models. With CoT, almost every name-error contrast meets our equivalence bound, although some accuracy effects remain. We then compare value readouts with excess name errors. The readout–error relationship depends on the low-accuracy exception, whose generated-chain probes also fail calibration. The behavioral reduction is clearer than its internal explanation.

2 Study design

We use Kudo et al.’s single-digit arithmetic task: equations with addition or subtraction, followed by a query. Our main level requires two operations, with the queried variable defined before its dependency. Two models are also tested on all five levels (Appendix A). Each problem has matched versions differing in one variable name (Table 1). A congruent number word denotes that variable’s true value; an incongruent word denotes a different digit. We vary the queried and intermediate variables separately. The name-suggested value cannot equal any variable’s value or input digit. Names occupy one token in every context, keep-

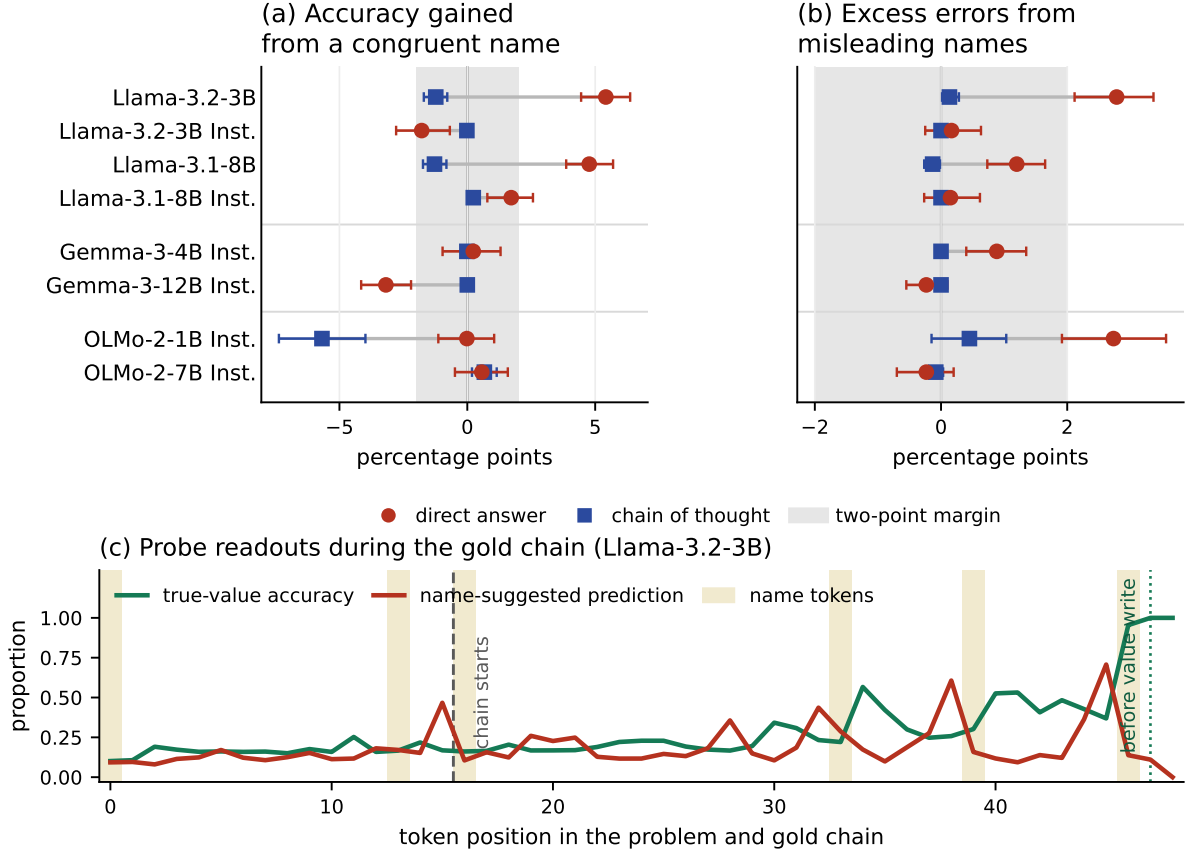


Figure 1: Eight panel models, queried-variable effects at level 3: **(a)** congruent-name accuracy gain and **(b)** excess name errors, averaged over three demonstration sets with 95% intervals. Grey marks the two-point margin. **(c)** Maximum-over-layers true-value accuracy and name-suggested prediction rate at each token position, pooled under a gold chain. The value readout concerns the queried variable. Shading marks name positions; the dotted line marks the position before its value.

Name	Problem (answer: 6)
Neutral	pen=1 + cup, cup=2 + 3; pen=?
Congruent	six=1 + cup, cup=2 + 3; six=?
Incongruent	four=1 + cup, cup=2 + 3; four=?

Table 1: Renaming the queried variable changes one cue while preserving the arithmetic. The incongruent name denotes 4, which appears nowhere else in the problem.

ing twins’ token positions identical.

Models either write a chain that restates and evaluates each equation or answer directly. Every behavioral comparison uses three fixed sets of three worked examples. We evaluate thirteen checkpoints from the Llama, Gemma and OLMo families; the model list and tuning comparisons are in Appendices A and F. The main behavioral table follows the preregistered 90% mean neutral-CoT accuracy floor; the excluded model’s results

remain in the appendix.

Our main measure is *excess name errors*: the rate of answers matching the misleading name’s digit, minus the neutral twin’s rate of answering that same digit. Subtracting this baseline matters when accuracy is low. We also report congruent-minus-neutral accuracy (*facilitation*) and neutral-minus-incongruent accuracy (*interference*). We call an effect reliable when its sign agrees across all three demonstration sets and its 95% cluster-bootstrap interval excludes zero. We call an effect small enough to meet the equivalence bound when its 90% interval fits wholly inside ± 2 points. We resample matched problems with their seed replicates together (Appendix B).

A linear probe predicts a variable’s digit from a hidden state at one token and layer. Probes train on separate neutral problems, with three training seeds. A control assigns a random digit to each name, testing how well the same probe can learn

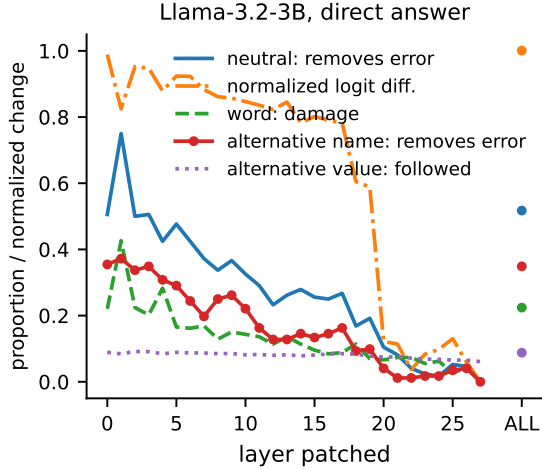


Figure 2: Llama-3.2-3B without CoT, queried variable: name errors removed by the neutral-twin patch, alongside neutral-word damage and the alternative-name control, by layer.

name identity (Hewitt and Liang, 2019). Patching replaces activations at the name’s tokens with a matched twin’s (Zhang and Nanda, 2024). We report both a neutral-word control for disruption and an alternative-name control for name-specific effects. Most internal measurements supply the correct chain token by token. We also train on generated neutral chains, selecting layers on computation-disjoint validation problems before testing held-out generations (Appendix N).

3 Results

3.1 Writing a chain limits name errors

Table 2 shows the naming effects at the main level. Without CoT, a congruent name raises Llama-3.1-8B’s accuracy by +4.8 points, and a misleading name produces +1.2 points of excess name errors on the queried variable. Across levels, the largest effects occur where direct-answer accuracy is near chance; the matched baseline prevents those arbitrary digit answers from being attributed to the name.

With CoT, 43 of 44 name-error contrasts meet the equivalence bound. The exception is OLMo-2-1B-Instruct: its intermediate-variable excess is +3.2 points [+2.5, +3.9]. Its neutral CoT accuracy is 47%, compared with 77% under letter names. CoT therefore limits the name’s numeric influence without removing every naming effect: 15 accuracy contrasts remain reliable (Appendix B).

3.2 Value readouts and task competence

A name’s digit can be readable without representing the computed value. For Llama-3.2-3B, the intermediate value is partly decodable after its defining equation (neutral accuracy 0.49; selectivity 0.32). At the query, the name-identity control scores 0.97. By the value-writing step, true-value accuracy reaches 0.98 or more and name-suggested predictions fall to zero. A Gemma token sweep also finds different readout timing for queried and intermediate values (Appendix J).

Across 9 models and 18 variable pairs, the 17 pairs with neutral probe accuracy of at least 0.98 have excess name errors within 0.4 points of zero. The exception’s intermediate readout is 0.22. Yet the descriptive correlation between model-mean misleading-name probe accuracy and excess name errors reverses from $r = -0.997$ to $+0.776$ when OLMo-2-1B is omitted (Appendix L). Overall task competence remains a competing account. Within-problem fits also disagree on the association’s sign (Appendix E).

Gold-trained probes score 19.4–62.2% on OLMo-2-1B’s neutral generated chains. Training on generated neutral chains raises this to 50.3–80.1%, still below the 90% calibration threshold in every cell. Llama-3.2-3B reaches 96.3–100.0% with the same design, but its name-error cases remain rare. The exception’s generated-chain readouts therefore remain descriptive (Appendix N).

Merely stating the correct value does not explain the pattern. We split the model’s generated chains by whether they state that value and compare only twins falling in the same split. The exception still has +3.6 points of excess name errors [+2.6, +4.7] across 2739 matched observations from 3 demonstration sets, despite stating the correct value. The full control is in Appendix D.

3.3 Patching localizes the no-chain effect

For Llama-3.2-3B without CoT, replacing name activations with the neutral twin’s removes 52% of name errors when all layers are patched (Figure 2). A span-by-layer sweep places the strongest effects at the defining equation in layers 0–7, where removal reaches 69% (Appendix K). The neutral-word control also damages 22% of correct answers. The patch identifies a region sensitive to the name; disruption limits claims about the restored computation.

The Llama-3.1-8B controls do not distinguish a

Model	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
			helps	errors	helps	errors
Gemma-3-12B-it	CoT	100 [100,100]	+0.0	+0.0	+0.0	+0.0
Gemma-3-12B-it	direct	78 [76,81]	-3.2*	-0.2	-3.7*	+0.4
Gemma-3-4B-it	CoT	100 [100,100]	-0.0	+0.0	-0.0	+0.0
Gemma-3-4B-it	direct	55 [48,59]	+0.2	+0.9*	+0.6	+0.3
Llama-3.1-8B-Instruct	CoT	100 [99,100]	+0.2	+0.0	+0.3	+0.0
Llama-3.1-8B-Instruct	direct	52 [45,65]	+1.7	+0.1	+5.2*	+0.2
Llama-3.1-8B	CoT	95 [86,100]	-1.3	-0.1	+3.8	-0.0
Llama-3.1-8B	direct	53 [41,62]	+4.8*	+1.2*	+3.8*	-0.0
Llama-3.2-3B-Instruct	CoT	100 [100,100]	-0.0	+0.0	-0.2	+0.0
Llama-3.2-3B-Instruct	direct	44 [40,50]	-1.8*	+0.2	+0.5	+0.5
Llama-3.2-3B	CoT	98 [97,100]	-1.2	+0.1	+0.9	-0.2*
Llama-3.2-3B	direct	22 [13,31]	+5.4*	+2.8*	+1.1	+1.8
OLMo-2-7B-Instruct	CoT	95 [93,97]	+0.7*	-0.1	-2.7	-0.0
OLMo-2-7B-Instruct	direct	54 [50,58]	+0.6	-0.2	+3.8*	-0.4

Table 2: Level 3, means over three demonstration sets (ranges in brackets). *Helps*: congruent-minus-neutral accuracy. *Errors*: excess name errors. Both are percentage points. * marks a consistent sign across sets and a 95% interval excluding zero. CoT name-error effects meet the two-point equivalence bound. The excluded low-accuracy model appears in Table 4; additional models and levels appear in Tables 7 and 9.

name-specific cause: neutral and alternative-name patches remove errors similarly. Under a gold CoT, injecting the misleading name into a neutral run of the exception model moves 2.7% of answers to the name-suggested value, while the word control changes 0.4% at the same layer. These interventions test only the selected models and positions; they do not establish a common causal mechanism across models.

4 Related work

Kudo et al. (2026) locate intermediate-value readouts during arithmetic CoT; our competing names distinguish a computed value from a lexical cue. Earlier Stroop studies test a single next token (Wang, 2026; Hu et al., 2026). Faithfulness studies perturb prompts, chains or parameters (Turpin et al., 2023; Lanham et al., 2023; Chen et al., 2025; Tutek et al., 2025); we combine a known conflicting cue with internal readouts. Positional copying and output priors can also explain arithmetic errors (Liu, 2026; Prabhakar et al., 2024); our copy control is reported in Appendix E. A companion code study (Anonymous, 2026) examines value writes in Python traces; this paper tests arithmetic name

errors and their association with value-step readouts.

5 Conclusion

Writing a chain limits answers suggested by misleading names in almost every tested arithmetic contrast. A correct value stated in the chain is insufficient to explain that reduction. The probe association depends on the low-accuracy exception, whose generated-chain probes fail calibration even after training on generated neutral chains. The behavioral finding therefore remains stronger than the proposed internal explanation.

Limitations

The task uses short synthetic arithmetic chains and English number words. It does not establish how misleading identifiers affect real code. Identifier-style names were tested at one level only (Appendix D). Models are at most 12B parameters and receive raw completion prompts; instruction-tuned models’ chat templates and long-form reasoning modes are untested. Demonstration choice changes accuracy substantially, which is why every behavioral claim must hold across three sets.

Linear probes reveal decodable information, not necessarily the representation used to generate an answer. Most internal measurements supply the correct chain token by token. Generated-chain probes trained on generated neutral chains meet the calibration threshold for Llama-3.2-3B but fail it for OLMo-2-1B (Appendix N). These probes cover two models and select layers on independent validation computations. The nine-model competence comparison is descriptive; variable roles are not independent models, and omitting OLMo reverses the readout-error correlation. It does not separate value availability from overall task competence. Patching changes mixed representations and can damage correct answers, limiting the causal conclusions. Finally, the chain repeats names and equations; the value-only format changes performance and cannot be treated as an equivalent intervention (Appendix H).

References

- Anonymous. 2026. Misleading identifiers change what code traces write. Under review.
- Akhiad Bercovich et al. 2025. Llama-Nemotron: Efficient reasoning models. *arXiv preprint arXiv:2505.00949*.
- Yanda Chen et al. 2025. Reasoning models don't always say what they think. *arXiv preprint arXiv:2505.05410*.
- Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A. Wichmann, and Wieland Brendel. 2019. ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations*.
- Gemma Team. 2025. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*.
- Aaron Grattafiori et al. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.
- John Hewitt and Percy Liang. 2019. [Designing and interpreting probes with control tasks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743.
- Xiaoyang Hu, Mike Angstadt, Shane Storks, Zan Huang, Aman Taxali, Alex Weigard, Richard L. Lewis, and Chandra Sripada. 2026. Conflict and congruency effects in large language models: In-weight and in-context competition in a verbal conflict task. *arXiv preprint arXiv:2608.11510*.
- Keito Kudo, Yoichi Aoki, Tatsuki Kuribayashi, Shusaku Sone, Masaya Taniguchi, Ana Brassard, Keisuke Sakaguchi, and Kentaro Inui. 2026. [LLMs faithfully and iteratively compute answers during CoT: A systematic analysis with multi-step arithmetics](#). In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 1114–1153.
- Man Ho Lam, Chaozheng Wang, Jen-tse Huang, and Michael R. Lyu. 2025. [CodeCrash: Exposing LLM fragility to misleading natural language in code reasoning](#). In *Advances in Neural Information Processing Systems*, volume 38, pages 133886–133932.
- Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, Kamilė Lukošūtė, Karina Nguyen, Newton Cheng, Nicholas Joseph, Nicholas Schiefer, Oliver Rausch, Robin Larson, Sam McCandlish, Sandipan Kundu, and 11 others. 2023. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*.
- Cuong Chi Le, Minh V. T. Pham, Cuong Duc Van, Hoang N. Phan, Huy N. Phan, and Tien N. Nguyen. 2025. When names disappear: Revealing what LLMs actually understand about code. *arXiv preprint arXiv:2510.03178*.
- Jack Le, Anh H. N. Nguyen, and Tien N. Nguyen. 2026. Do machines struggle where humans do? LLM and human comprehension of obfuscated code. *arXiv preprint arXiv:2606.31725*.
- Ming Liu. 2026. The readout shortcut: Positional number copying dominates arithmetic CoT readout in small language models. *arXiv preprint arXiv:2605.22870*.
- Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, and 2 others. 2019. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.
- Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12(85):2825–2830.
- Akshara Prabhakar, Thomas L. Griffiths, and R. Thomas McCoy. 2024. [Deciphering the factors influencing the efficacy of chain-of-thought: Probability, memorization, and noisy reasoning](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 3710–3724.

Skipper Seabold and Josef Perktold. 2010. [Statsmodels: Econometric and statistical modeling with Python](#). In *Proceedings of the 9th Python in Science Conference*, pages 92–96.

J. Ridley Stroop. 1935. Studies of interference in serial verbal reactions. *Journal of Experimental Psychology*, 18(6):643–662.

Team OLMo. 2024. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*.

Miles Turpin, Julian Michael, Ethan Perez, and Samuel Bowman. 2023. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems*, volume 36, pages 74952–74965.

Martin Tutek, Fateme Hashemi Chaleshtori, Ana Marasović, and Yonatan Belinkov. 2025. [Measuring chain of thought faithfulness by unlearning reasoning steps](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 9946–9971.

Han-yu Wang. 2026. Persistent priors, preserved targets: A Stroop-style paradigm for lexical override. *arXiv preprint arXiv:2606.07555*.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, and 3 others. 2020. [Transformers: State-of-the-art natural language processing](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online. Association for Computational Linguistics.

Fred Zhang and Neel Nanda. 2024. Towards best practices of activation patching in language models: Metrics and methods. In *International Conference on Learning Representations*.

A Task, models and replication

A problem contains single-digit variables, addition or subtraction, and a query. The five levels vary dependency order, depth and the presence of an unused equation (Table 3). Three same-level worked examples precede each test problem. The evaluated models are Gemma-3-12B-it, Gemma-3-4B, Gemma-3-4B-it, Llama-3.1-8B, Llama-3.1-8B-Instruct, Llama-3.2-3B, Llama-3.2-3B-Instruct, OLMo-2-1B-Instruct, OLMo-2-7B-Instruct, plus four derived checkpoints (Appendix F).

We reimplement Kudo et al. (2026)’s task and probe recipe rather than using their code. The

Level	Problem (neutral names)	Steps	Held	Distr.
1	pen=1 + cup, cup=2; pen=?	1	1	0
2	pen=2 + 3, cup=1 + pen; cup=?	2	0	0
3	pen=1 + cup, cup=2 + 3; pen=?	2	1	0
4	pen=1 + cup, cup=2 + 3, box=4 + 5; pen=?	2	1	1
5	pen=1 + cup, cup=2 + box, box=1 + 2; pen=?	3	2	0

Table 3: The five levels of Kudo et al. (2026): additions or subtractions needed (*Steps*), variables unresolved when first read (*Held*), and unused equations (*Distr.*). At level 1 the intermediate value is stated rather than computed.

shared choices are single-digit outputs, same-level demonstrations, a 10-way linear residual-stream classifier, full-batch SGD at 10^{-3} for 10,000 epochs, and their span-by-layer-window patching design. For replication we train on letter names and report t^* , t_{eq}^* , $\text{Acc}^{<\text{CoT}}$ and $\text{Acc}^{>\text{CoT}}$ at $\tau = 0.9$. The Llama-3.2-3B value-writing positions agree with their Table 3: equation 5 for the queried variable and 2 for the intermediate. Our additions are the conflicting names, matched baselines, demonstration-set claim rule and equivalence bound.

On the letter condition, pre-chain probe maxima for Llama-3.2-3B are 30.4 and 47.9, compared with 17.8 and 33.2 reported by Kudo et al. (2026). The value-writing positions agree; the accuracy difference may reflect training details. Llama-3.1-8B’s intermediate value becomes decodable at equation 2, but its queried value becomes decodable at equation 3, earlier than Llama-3.2-3B’s equation 5. The larger model can therefore expose that value before writing its final evaluation. Full heatmaps appear in Appendix K.

B Equivalence checks

Following preregistration Amendment 3, equivalence requires a 90% cluster-bootstrap interval wholly inside the ± 2 -point margin (two one-sided 5% tests). We resample matched sets with their demonstration-seed replicates together, using 4,000 bootstrap draws. Other behavioral intervals remain at 95%. For CoT accuracy, 61 of 88 facilitation and interference contrasts meet the same equivalence bound. Of the 15 reliable accuracy contrasts, 5 are also equivalent within the margin and 10 are not; a detectable effect can still be small. The written-value control uses the same matched pairs for its excess estimate and interval; the ex-

Model	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
			helps	errors	helps	errors
OLMo-2-1B-Instruct	CoT	47 [44,51]	-5.7*	+0.4	+7.0*	+3.2*
OLMo-2-1B-Instruct	direct	34 [30,38]	-0.0	+2.7*	+3.0*	-0.2

Table 4: The level-3 model below the preregistered 90% mean neutral-CoT accuracy floor, excluded from the main behavioral table. Columns and claim markers follow Table 2.

ception model’s estimate pools all three demonstration sets and clusters on matched set.

Contrast	Cells	Equivalent	Reliable
CoT name errors	44	43	3
CoT accuracy	88	61	15
Direct name errors	44	31	13

Table 5: Equivalence within the two-point margin uses 90% intervals; reliability uses 95% intervals excluding zero and the same sign across all three demonstration sets. Each accuracy cell is one facilitation or interference contrast.

C Tokenizer checks

We test candidate names in five tokenizer contexts and retain words that form one token without merging with a neighbor. For Llama 3, all ten number words and 95 of 112 neutral candidates pass. Gemma and OLMo use the same lists; run-time checks confirm equal token lengths and positions within matched groups. Spaces around binary operators prevent merges such as 1-two. No instance in the reported runs is excluded by the layout check.

D Additional naming and written-value controls

Written values. We split generated chains by whether they state the target’s true value. An observation enters the excess estimate only when its neutral twin falls in the same split. The estimate and its 95% interval use those same paired observations. Pooled over eligible cells, the excess is +0.08 points in 132 cells where the value is stated and -0.10 in 27 where it is not. The exception persists in the stated-value split (main text), so this split does not explain its effect.

Identifier-style names. At level 3, four models receive names such as q4 instead of four, under three demonstration sets. Without CoT, queried-variable excess name errors range from -0.2 to

+0.2 points; 0 of 4 cells are reliable. For number words on the same models, the range is -0.2 to +2.8, with 3 of 4 reliable. The numeric name-error effect is therefore stronger for number words in this comparison; congruent identifier names can still improve accuracy.

Word-class effects. With CoT at level 4, a number word on the queried variable costs Llama-3.2-3B 9.9 accuracy points whether the name agrees with the value or not. Those errors restate the wrong equation. This is distinct from answering the digit the name denotes, which is why the name-suggested value result cannot be read as removal of every naming effect.

Unused variables. A number word on the level 4 distractor changes accuracy by at most 1.2 points. Its no-CoT excess name errors range from +2.1 to +2.3; 2 of 2 cells are reliable. With CoT, the range is +0.0 to +1.0, with 0 of 2 reliable. At the answer position, probe probability of the name-suggested value is 0.004 for a computed variable and 0.118 for the distractor; the latter’s true value has probe accuracy 0.04.

E Full sweeps and selectivity

Mixed-effects check. We compare the paired bootstrap with a mixed logit model, $\text{name_error} \sim \text{incongruent} + C(\text{seed})$, with a random intercept for matched set. Each twin’s outcome refers to the same name-suggested digit. Both procedures use the same fixed subsample of 600 sets per cell. Of 57 fitted cells, 50 agree on whether the interval excludes zero; 11 additional cells have no outcome variation. The mixed fit supports an effect in 5 cells where the bootstrap does not and withdraws support in 2 cells where it does (Gemma-3-12B-it, value-only chain, queried variable; Llama-3.1-Nemotron-Nano-8B, CoT, intermediate variable). Subsampling also removes support from 1 starred Table 2 cell under both procedures (Gemma-3-4B-it, no CoT, queried variable). The mixed model

Target	Read	Errors	Neutral removed	Word dmg.	Alt. removed	Alt. followed
v1	value	0	—	1.6	—	0.0
v1	answer	0	—	1.6	—	0.0
v2	value	28	92.9	7.3	82.1	4.0
v2	answer	3	100.0	0.8	100.0	0.6

Table 6: OLMo-2-1B-Instruct, CoT, 500 matched pairs per target, patching name tokens at all layers. Removed: percentage of baseline name-suggested predictions removed by the neutral-twin patch. Word damage: percentage of correct neutral predictions damaged by the word control. Alt. removed: baseline name-suggested predictions removed by the alternative-name patch. Alt. followed: percentage following that source’s name-suggested value. Rates are percentages; — means undefined. These readouts force the gold chain, so their baseline errors differ from errors in the model’s own generated chains.

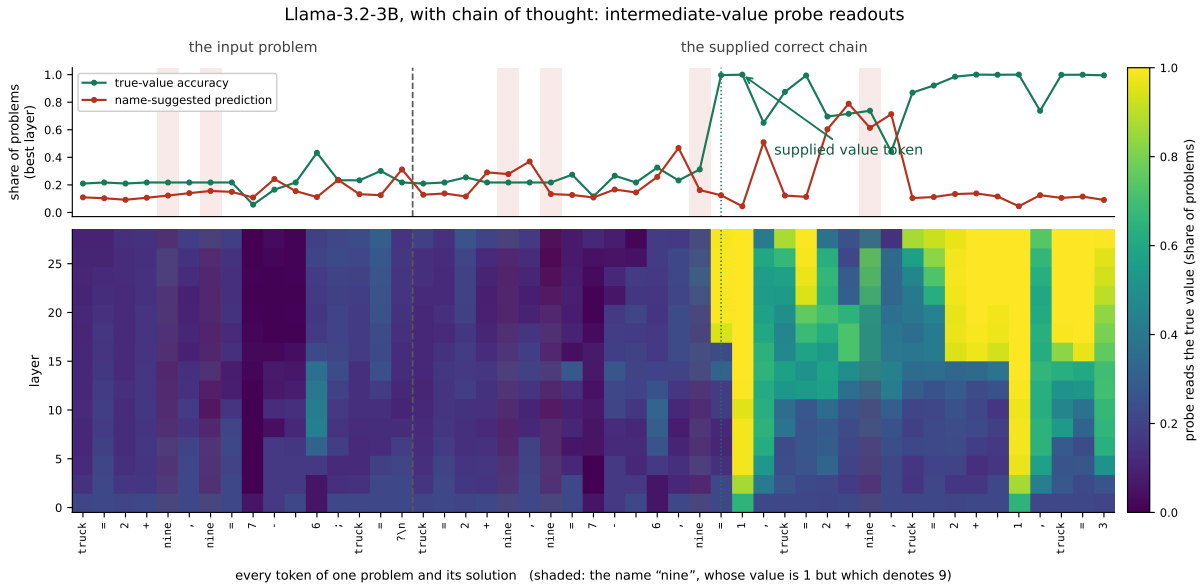


Figure 3: In the layout of Kudo et al.’s Figure 2. Per-token probes for the incongruent condition, Llama-3.2-3B, level 3, CoT. Top: how often the best-layer probe reads the true value (green) or the name-suggested value (red). Bottom: true-value accuracy by layer. The displayed tokens are from one example; the rates pool problems with different values and names. Dashed line: start of the supplied correct chain; dotted line: position before the target value; arrow: the supplied value token. Shading marks the name’s tokens.

uses variational Bayes, so its interval is an approximate posterior interval. We retain the full-data bootstrap as the primary estimate on the reported percentage-point scale.

Instance-level association. Within each model, level, regime, variable and position, we regress name errors on the probe margin, $\log p_{\text{true}} - \log p_{\text{name}}$, where p_{name} is the probe probability of the name-suggested value, with standard errors clustered on matched set. The layer is selected by neutral accuracy. Of 63 fits, 29 have $p < .05$: 24 coefficients are negative and 5 positive. The direction is not consistent, and 4 positive results come from the model supplying most fits. This leaves the instance-level association unresolved.

The model-level pattern is also sensitive to panel composition (Appendix L).

Name-error share. For comparison with shape–texture studies (Geirhos et al., 2019), we also divide name errors by the total of name errors and correct answers, excluding congruent trials. Across 18 level 3 model–variable pairs, the index ranges from 0.02 to 0.38 without CoT and from 0.00 to 0.16 with CoT. The main text uses paired excess because the index does not subtract how often the neutral model answers that digit.

Positional-copy control. Across 4110 generated CoT name errors over models, levels and demonstration sets, the number immediately pre-

ceding the answer is the name-suggested value in 13% of cases and the target’s true value in 11%. A trailing name-suggested value is therefore absent from many errors. This check constrains the positional-copy explanation of Liu (2026), but does not identify what was copied in each case.

Probes report true-value accuracy, name-suggested prediction rate, probability of the name-suggested value and margin, with name-identity control accuracy and selectivity (Hewitt and Liang, 2019). P1 is the name token, P2 the end of its defining equation, P3 the query, P4 the position before its value is written, and P5 the position before the final answer. P1 is a reference, not evidence for a computed value. At P3 the control is near 1, so a readout of the name-suggested value there may reflect name identity. Figure 3 shows the CoT token sweep; Appendix K contains the other conditions and direct regime. Letter-trained replication uses the metrics in Appendix A.

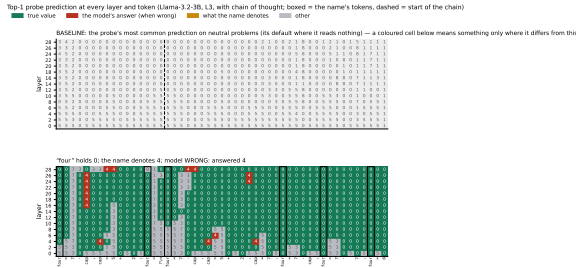


Figure 4: In the layout of Kudo et al.’s Figure 3, for problems this model answers with the name-suggested value. Each cell is the probe’s top-1 prediction at that layer and token: green the variable’s true value, red the value the model answered, amber the value its name denotes, grey anything else. The first panel is the probe’s most common prediction on neutral problems, which is what it emits where it reads nothing; cells carry information only where they differ from it.

F How the models were tuned

We compare six checkpoints on the Llama-3.1-8B architecture: pretrained, instruction-tuned, single-task LoRA, multi-task LoRA, TIES-merged adapters and reasoning-tuned. Adapters are folded into the weights before evaluation. Shared tokenizers and identical matched problems permit paired comparisons (Table 7). A pretrained/instruct Gemma pair supplies a second family comparison.

The pretrained and reasoning-tuned Llama checkpoints have reliable positive queried-variable excess name errors without CoT (+1.20

Tuning	Model	Queried	Intermediate	Acc.
base	Llama-3.1-8B	+1.20*	-0.02	53.4
instruct	+ instruct	+0.15	+0.22	52.2
single-task finetune	+ task LoRA	+0.05	+0.23	52.2
multi-task finetune	+ multi-task LoRA	-0.58*	-0.08	46.1
weight merge	+ TIES merge	-0.10	+0.10	51.9
reasoning-tuned	Nemotron-Nano-8B	+1.35*	+0.33	39.4
base	Gemma-3-4B	+0.08	+0.02	60.2
instruct	+ instruct	+0.88*	+0.33	54.9

Table 7: Level 3, no CoT, name errors above the matched twin’s rate, mean over three demonstration sets. Acc. is neutral accuracy. * as in Table 2. The bottom block repeats the pretrained-to-instruct step in a second family, Gemma-3-4B.

and +1.35 points). The three task-tuned variants differ from the instruction-tuned checkpoint by at most 0.73 points across 6 comparisons. These are observed differences, not tests of checkpoint-to-checkpoint equivalence. Every checkpoint’s CoT excess is at most 0.20 points in magnitude.

Task accuracy complicates interpretation. The multi-task model’s reliable negative excess (-0.58) accompanies lower neutral accuracy (46.1%, versus 52.2%). The reasoning-tuned checkpoint has the lowest accuracy (39.4%). It also requires a system instruction to enable long-form reasoning; our raw-completion prompts leave that mode off. The result concerns the checkpoint under this prompt, not reasoning mode in deployment.

Only the instruct Gemma effect passes the claim rule (+0.88, versus +0.08 pretrained). The difference is untested and both estimates lie within the margin, so the pair cannot establish where in training name interference arises.

G Probe recipe and patching scope

We check two implementation choices inherited from Kudo et al. (2026): the probe optimizer and the name occurrences included in a patch.

Recipe. We refit the level-3 probes for Llama-3.2-3B with L-BFGS, and with SGD on standardized inputs, against the reported recipe.

Position	Variable	SGD (reported)	L-BFGS	standardized
defining equation	queried	0.184	0.190	0.183
defining equation	intermediate	0.496	0.422	0.420
query	queried	0.183	0.316	0.280
query	intermediate	0.271	0.366	0.393
value step	queried	1.000	1.000	1.000
value step	intermediate	0.982	0.962	0.972
answer	queried	1.000	1.000	1.000

Table 8: Best-layer accuracy for the true value, neutral instances, under three probe recipes.

All three recipes give the same qualitative re-

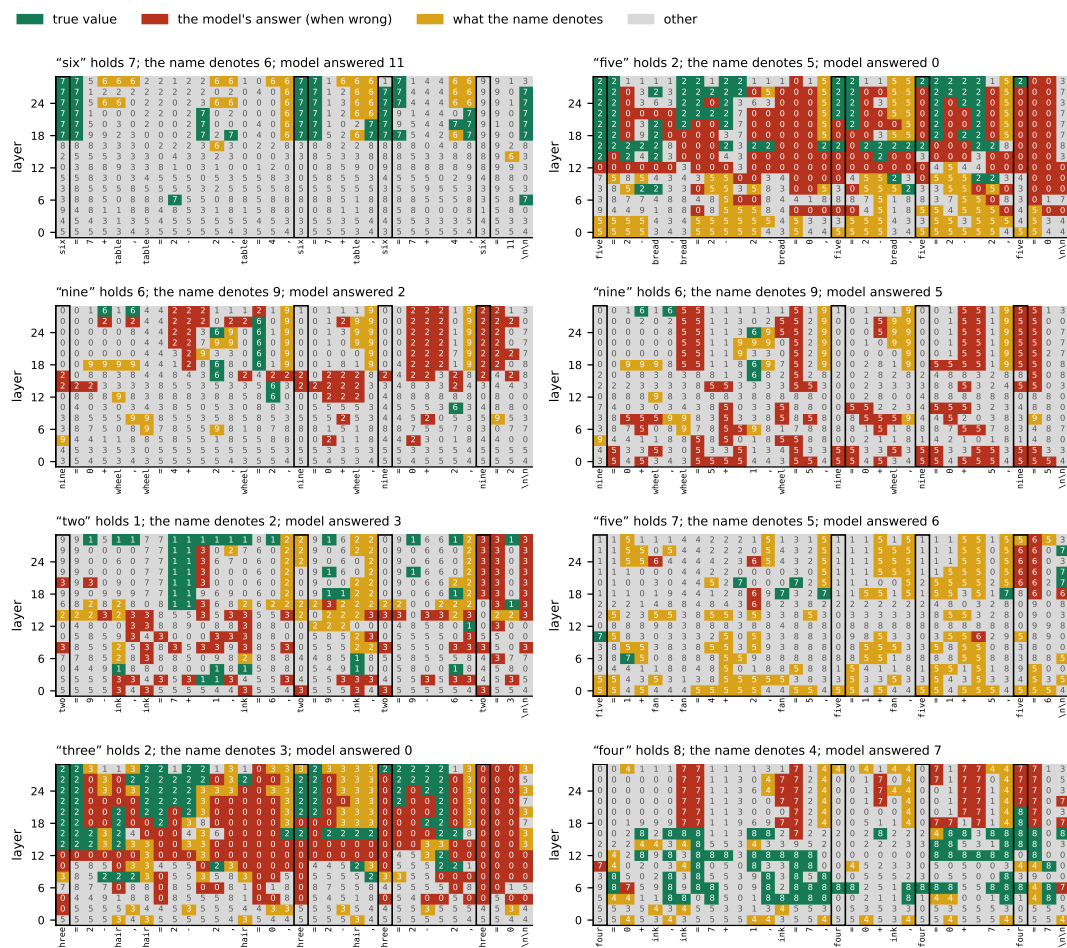


Figure 5: Probes trained at the value step, applied to the model’s own generated chain rather than to a force-decoded gold chain, for the 8 level-3 incongruent problems that Llama-3.2-3B answered wrongly under CoT (the first 8 of 108 wrong answers in 2,000 instances). Probes are trained on neutral problems at the step that writes the variable’s value and applied at every generated token and every second layer; colours as in Figure 4, boxed columns the name’s own tokens. This small sample contains no name errors and is illustrative rather than an estimate of their rate. The fixed-sample transfer analysis and separate additional name-error cases appear in Appendix M.

sult: the value is decodable where the chain writes it (0.96–1.00) and not at the end of its defining equation (0.18–0.50). The query position moves, from 0.271 under the reported recipe to 0.393 under standardization, within a range of 0.18–0.39 over recipes and variables. We already report that position as the presence of the name rather than a decoded value, because its control task scores 0.97, so this variation does not change the interpretation of that position.

Scope. Patching every occurrence of the name includes the copy the model writes in its own output. In the direct regime, restricting the patch to the name’s occurrences in the prompt leaves the intermediate variable essentially unchanged, since its name never appears in the answer: removal

is 43% at layers 0–3 under both scopes. For the queried variable the best windows remove similar shares (52% at layers 4–7 prompt-only, against 53% at layers 0–3), the prompt-only patch removes less in the later windows, and the two disagree most in the first window, where the prompt-only patch removes 3% and damages 76% of correct answers. That is mechanical: the queried variable’s name also sits at the position we read, so patching its prompt copies alone leaves the model reading a name whose early representation no longer matches. The localization we report does not depend on the choice.

H A chain that writes values without equations

Our chain restates each equation before substituting into it, as Kudo et al.’s does. Their Tables 5–6 also report a format that writes only the sub-results, cup=5, pen=6. That format is the arithmetic counterpart of a code trace, which writes a variable’s value and never the expression that produced it, so we ran it on the panel at level 3 with three demonstration sets, with two predictions registered in advance for both Llama models: that the excess name errors at the intermediate’s value step would exceed the 2-point margin, and that the value would still be decodable there (probe accuracy at least 0.85), making any failure one of readout.

Both predictions fail as registered. Llama-3.1-8B writes the name-suggested value at the intermediate’s value step +0.15 points above its twin, inside the margin. Llama-3.2-3B is at +9.20, which passes our claim rule, but across the three demonstration sets that is +1.7 / +24.2 / +1.7, so one set carries most of it. More informative is what the probes say. At the step that writes the value, the true value is decodable at 0.81 and 0.59 for Llama-3.1-8B and at 0.61 and 0.79 for Llama-3.2-3B, against 1.00 and 0.99 for the same model under the full chain. Neutral accuracy falls with it, to 54% and 31%.

Removing the equations lowers both accuracy and true-value probe readouts. It therefore fails to isolate a readout error while preserving those measures. This result does not support a shared explanation for value-only arithmetic chains and code traces.

I Additional levels, models and demonstration sets

Table 9 repeats Table 2 for the other four levels. The pretrained Gemma-3-4B, the base half of the Gemma pair, is reported in Table 7 rather than in Table 2.

Data, compute and software. Each level has 2,000 matched sets, each rendered in every condition, and 10,000 separately generated problems for training probes; every SGD probe is trained with 3 seeds. Every run here is inference and the probes are linear; the three adapter checkpoints of Appendix F were trained in a separate project, and that compute is not counted below.

The recorded allocation time totals at most 185 GPU-hours on NVIDIA A30, H100 and H200 cards, including idle time and shared workers that also ran companion-code experiments. The models are from the Llama 3 (Grattafiori et al., 2024), Gemma 3 (Gemma Team, 2025) and OLMo 2 (Team OLMo, 2024) families, with the reasoning-tuned checkpoint from Bercovich et al. (2025), each used under its release licence for research. Models run in PyTorch (Paszke et al., 2019) with Transformers (Wolf et al., 2020), and the probes are trained in PyTorch; the L-BFGS refit of Appendix G uses scikit-learn (Pedregosa et al., 2011), and the mixed-effects check uses statsmodels (Seabold and Perktold, 2010).

Model	Level	Regime	Neutral accuracy (%)	number word on the queried variable		number word on the intermediate variable	
				helps	errors	helps	errors
Llama-3.1-8B	1	CoT	99 [98,100]	+0.0	+0.0	+0.3	+0.0
Llama-3.1-8B	1	direct	74 [38,99]	+16.1*	+11.2*	+8.8*	+1.3
Llama-3.2-3B	1	CoT	97 [95,100]	+1.1	+0.1	-2.4	+0.3
Llama-3.2-3B	1	direct	46 [21,93]	+0.6	+5.9*	-3.9*	+2.6*
Llama-3.1-8B	2	CoT	100 [100,100]	-0.1	+0.0	-0.0	+0.0
Llama-3.1-8B	2	direct	36 [33,37]	+11.8*	+0.7	+2.6	+2.2
Llama-3.2-3B	2	CoT	94 [90,98]	+3.7	+0.5*	+1.0*	-0.1
Llama-3.2-3B	2	direct	29 [27,30]	+10.8*	+2.5*	+2.6*	+2.2
Llama-3.1-8B	4	CoT	88 [63,100]	+1.0	-0.0	-2.3	+0.2
Llama-3.1-8B	4	direct	28 [26,31]	+13.7*	+5.8	+6.0*	+0.2
Llama-3.2-3B	4	CoT	64 [26,85]	-10.2*	+0.7	-2.6*	+0.4
Llama-3.2-3B	4	direct	20 [16,23]	-1.6	+1.9	-0.2	+1.2*
Llama-3.1-8B	5	CoT	65 [43,100]	+1.2	-0.3	+0.1	-0.1
Llama-3.1-8B	5	direct	20 [13,26]	+20.2*	+20.5*	+0.9	+0.4
Llama-3.2-3B	5	CoT	69 [48,97]	-1.4	+0.1	-2.5	-0.1
Llama-3.2-3B	5	direct	12 [11,12]	+4.7*	+3.5*	+1.1*	+0.5

Table 9: As Table 2, for levels 1, 2, 4 and 5 (Llama-3.2-3B and Llama-3.1-8B). Level 1 is the no-computation control: the intermediate variable’s value is stated rather than computed.

Per-set values, the pooled estimate and the claim rule are released with the code, together with the level-4 irrelevant-name-suggested value control, the name-suggested value-distance analysis, the teacher-forced name-suggested prediction rates, and the error taxonomy behind the word-class effect.

J Additional value readouts

We extend the value-step comparison to Llama-3.2-3B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-12B-Instruct. Each probe trains on 10,000 separate neutral problems with three probe seeds, using demonstration set 7. Table 10 reports the added pairs; the main comparison includes them. We select layers by neutral selectivity, as in the original analysis. The misleading-name accuracy uses the same selected layer. This extension tests the aggregate association; it does not resolve the within-problem link.

Model	Variable	Layer	Neutral	Control	Misleading
Llama-3.2-3B-Instruct	queried	18	99.9	26.3	100.0
Llama-3.2-3B-Instruct	intermediate	19	99.9	29.5	99.9
Llama-3.1-8B-Instruct	queried	20	100.0	21.0	100.0
Llama-3.1-8B-Instruct	intermediate	20	100.0	25.7	100.0
Gemma-3-12B-it	queried	40	100.0	13.1	100.0
Gemma-3-12B-it	intermediate	42	100.0	11.9	100.0

Table 10: Additional probes at each variable’s value-writing position. Neutral and misleading columns are true-digit accuracy; control is neutral name-identity accuracy. Rates are percentages, averaged over three probe seeds. The layer maximizes neutral accuracy minus control accuracy.

We also probe every token for Gemma-3-4B-Instruct (Figures 6 and 7), using 4,000 independent neutral training problems and 1,000 test problems per condition. The tokens shown come from one example; the curves and heatmaps pool problems with different names and values. Maximum-over-layers curves describe where a digit can be read; they do not identify the representation used to answer. On neutral problems, the best pre-chain accuracy is 26.1% for the queried digit and 50.3% for the intermediate digit. The queried readout first exceeds 90% at token 24; the intermediate readout first exceeds 90% at token 13. Token zero is the first token of the chain; negative positions are in the prompt.

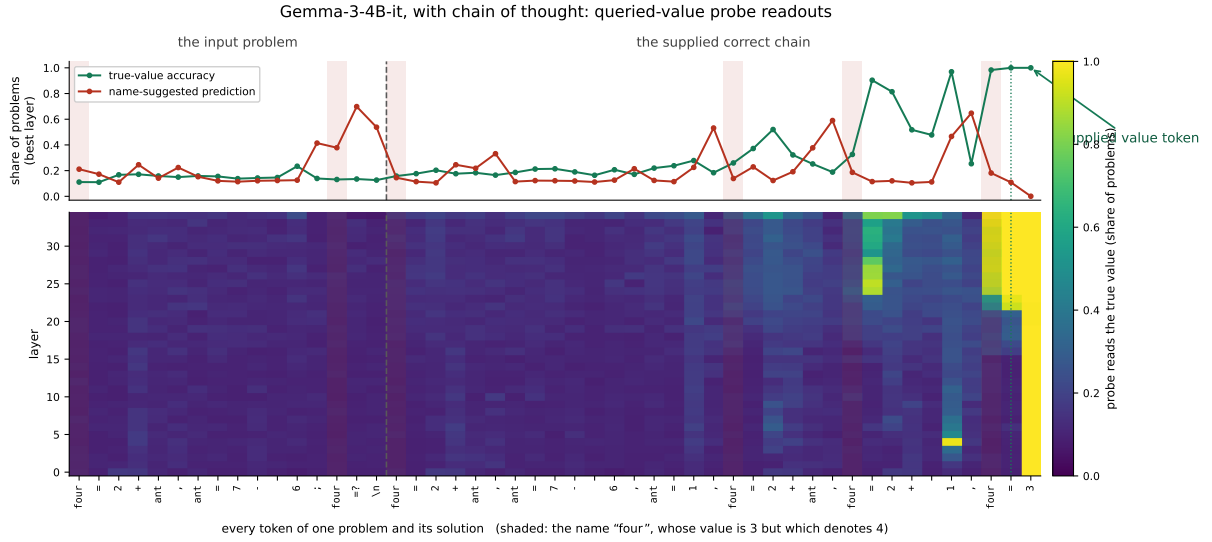


Figure 6: Gemma-3-4B-Instruct, queried variable, misleading-name problems under the correct chain. Top: maximum-over-layers true-value accuracy and name-suggested prediction rate; bottom: true-value accuracy by layer. Shading marks the variable's name; the dotted line marks the position before its value; the arrow marks the supplied digit itself.

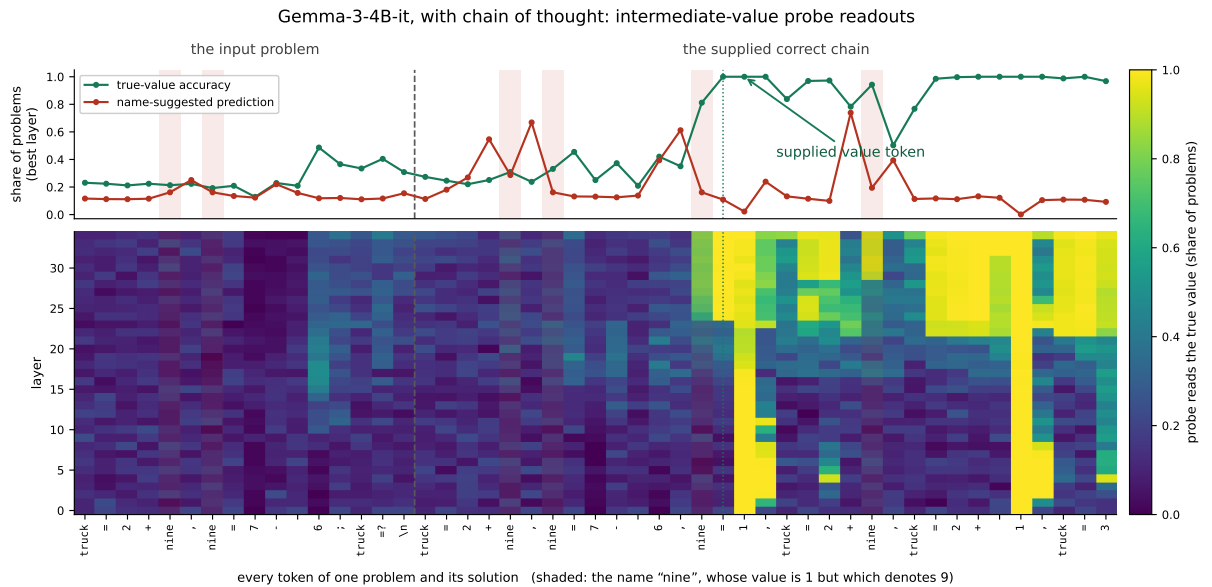


Figure 7: The same Gemma token sweep for the intermediate variable. Both figures use neutral-trained probes; no misleading-name problems enter training.

720
721
722

K Every figure at full size

Each figure below is reproduced as a full-width figure on a portrait page, within the ACL margins.

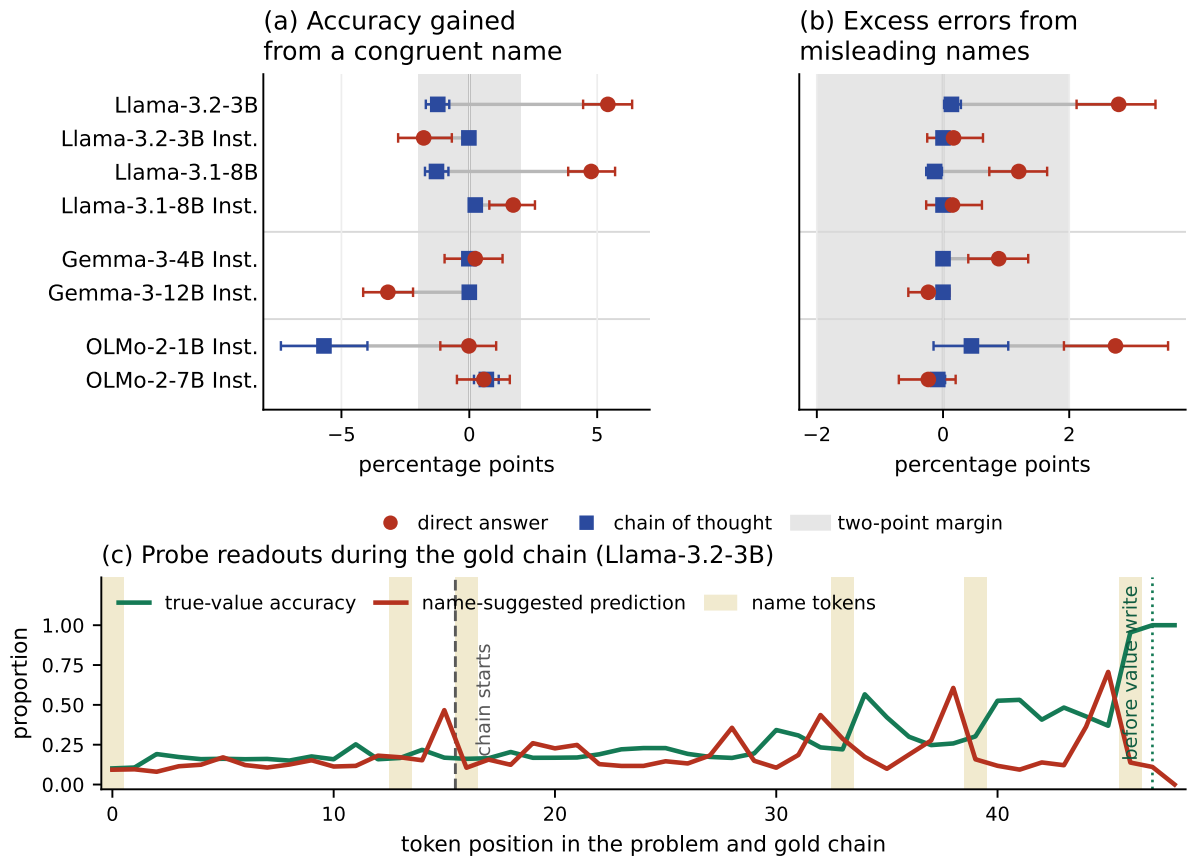


Figure 8: Figure 1 at full size.

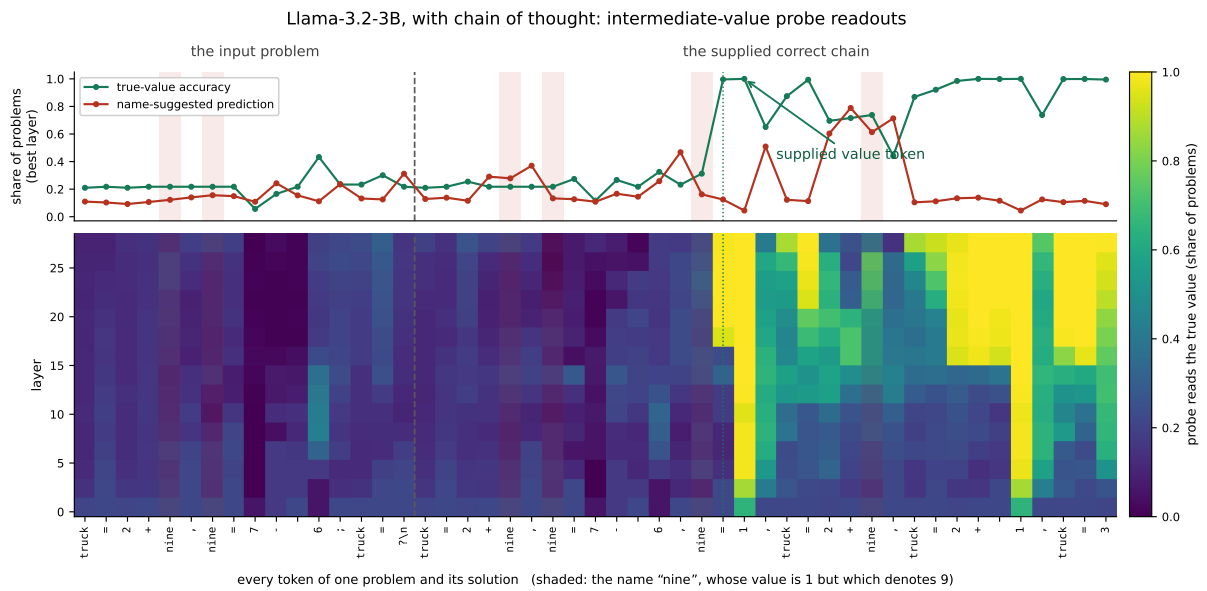


Figure 9: Figure 2 at full size: per-token probes, chain-of-thought regime, intermediate variable.

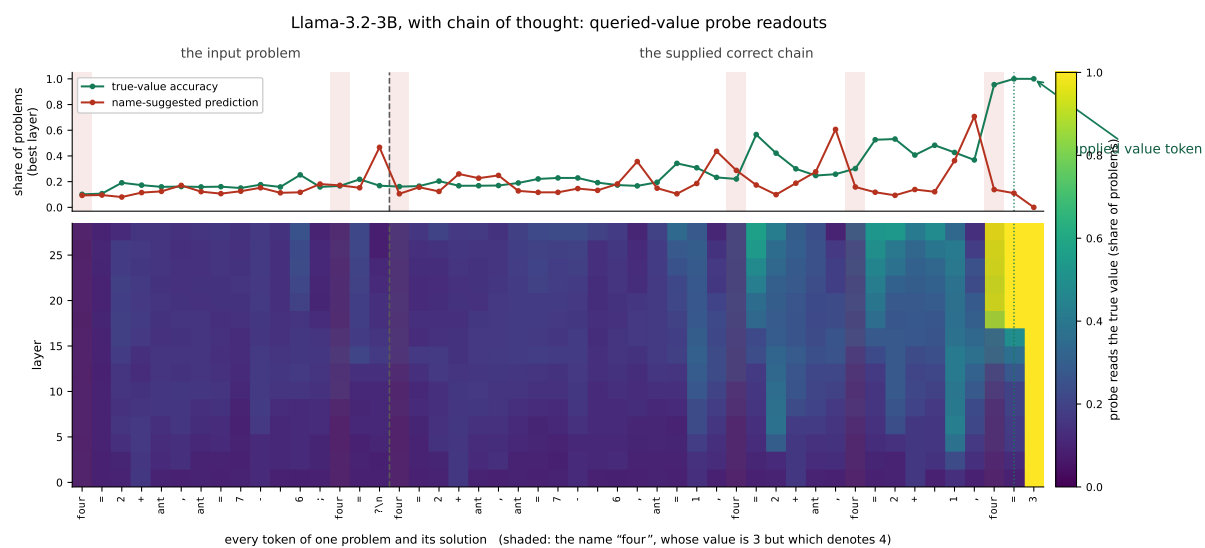


Figure 10: Per-token probes, chain-of-thought regime, queried variable.

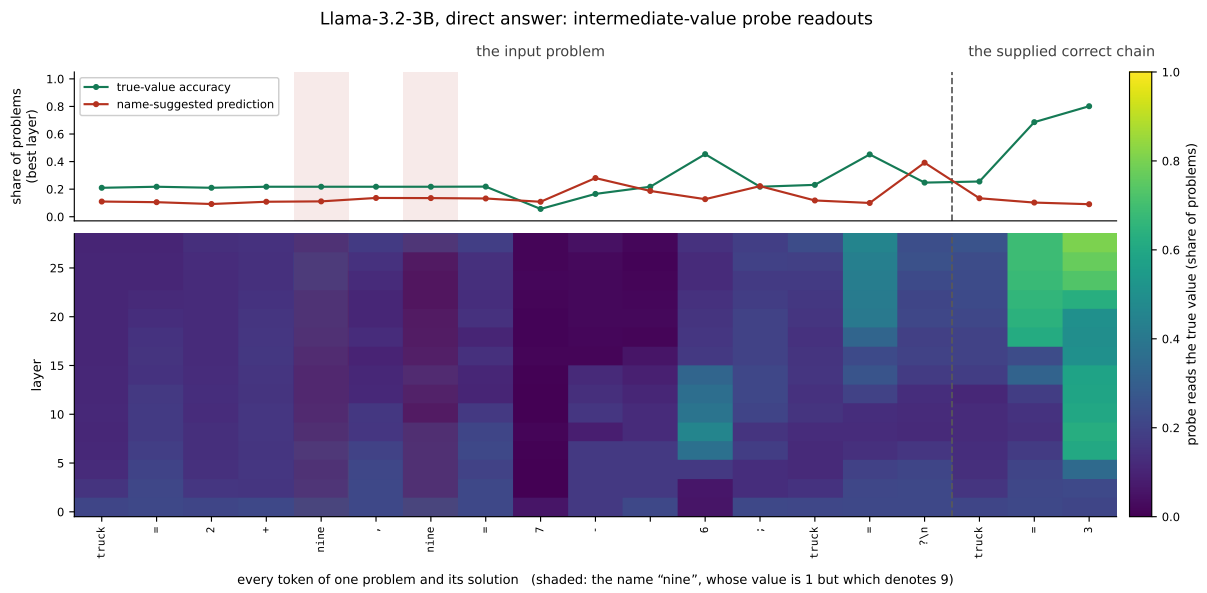
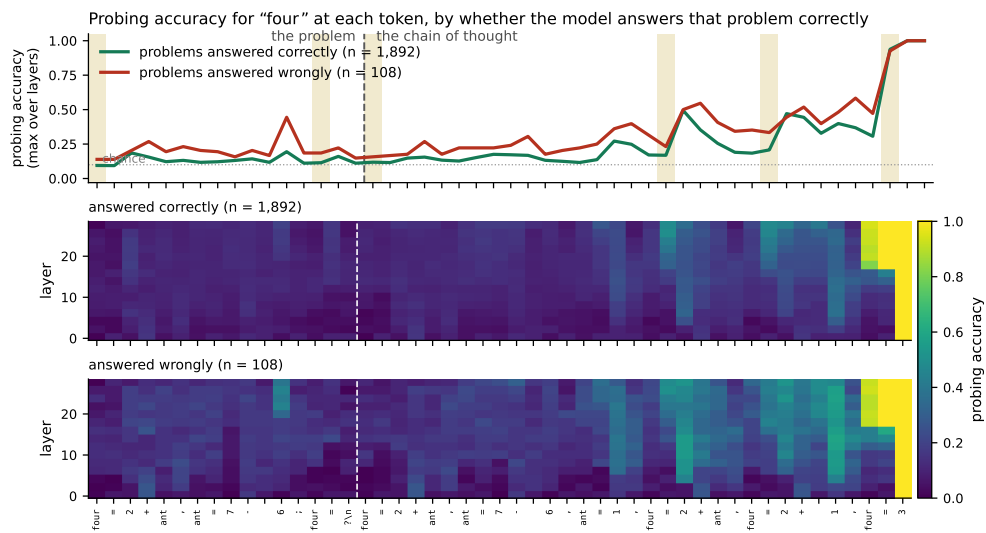


Figure 11: Per-token probes, direct regime, intermediate variable.



Probes read the variable's true value; states come from a forced pass over the correct chain, so the lower panel asks what the state holds for problems this model gets wrong unaided.

Figure 12: Probing accuracy split by whether the model answers the problem correctly on its own, with the layer sweep for each half.

Top-1 probe prediction at every layer and token (Llama-3.2-3B, L3, with chain of thought; boxed = the name's tokens, dashed = start of the chain)

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

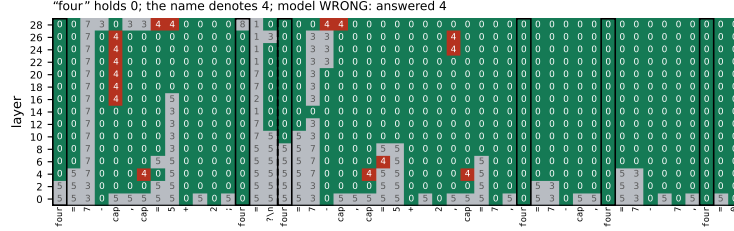
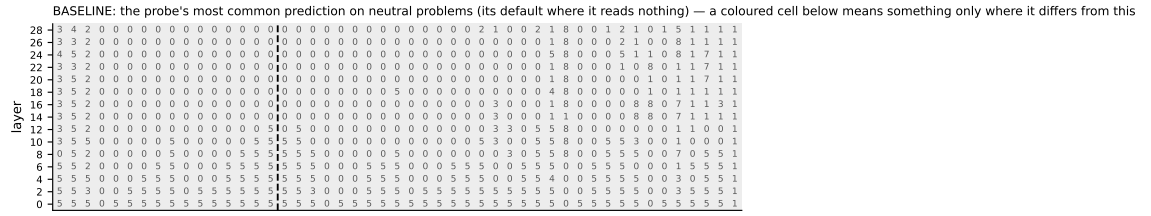


Figure 13: Top-1 probe prediction at every layer and token, on a problem answered with the name-suggested value. Layout after Kudo et al.'s Figure 3.

Top-1 probe prediction at every layer and token (Llama-3.2-3B, L3, with chain of thought; boxed = the name's tokens, dashed = start of the chain)

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

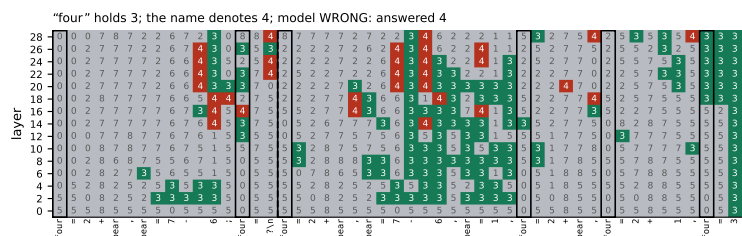
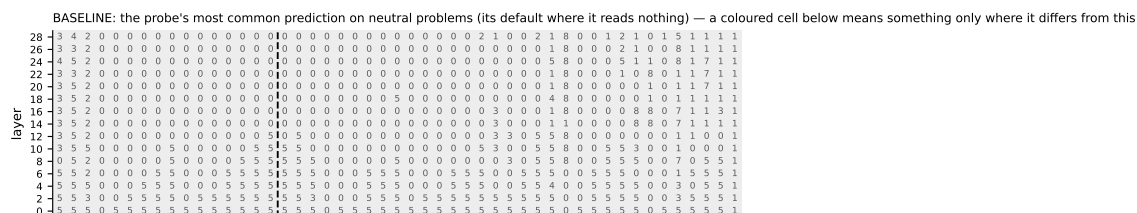


Figure 14: A second problem answered with the name-suggested value.

Top-1 probe prediction at every layer and token (Llama-3.2-3B, L3, with chain of thought; boxed = the name's tokens, dashed = start of the chain)

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

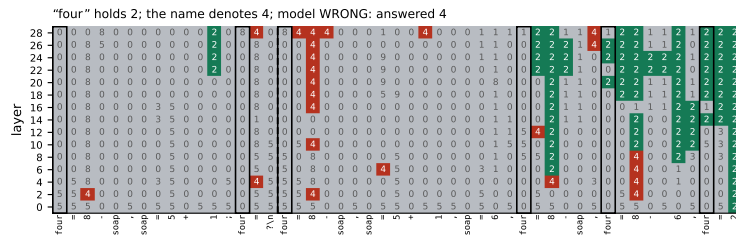
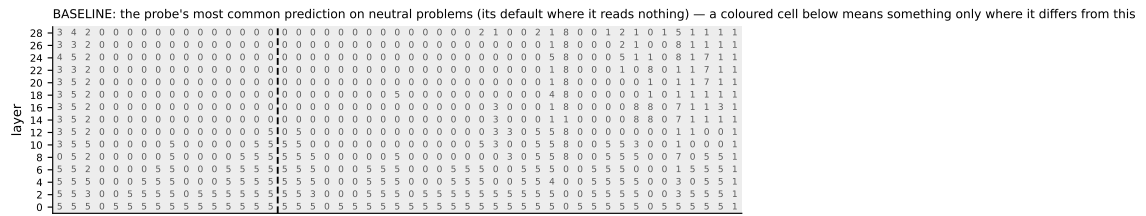


Figure 15: A third problem answered with the name-suggested value.

Top-1 probe prediction at every layer and token (Llama-3.2-3B, L3, with chain of thought; boxed = the name's tokens, dashed = start of the chain)

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

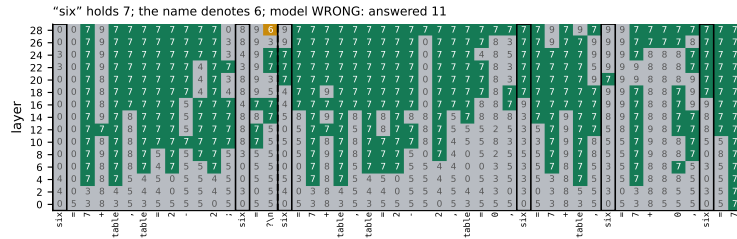
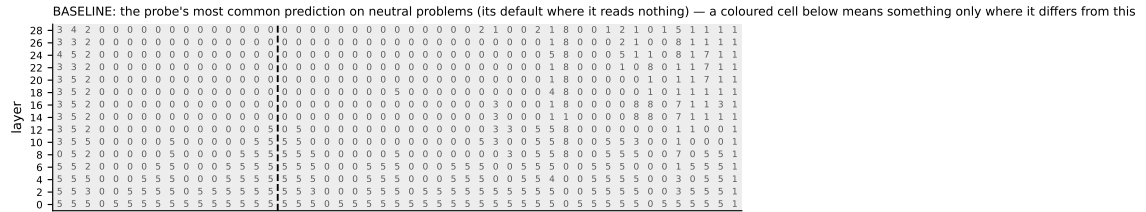
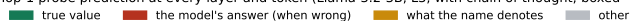


Figure 16: As above, on a problem answered wrongly for another reason.



BASELINE: the probe's most common prediction on neutral problems (its default where it reads nothing) — a coloured cell below means something only where it differs from this

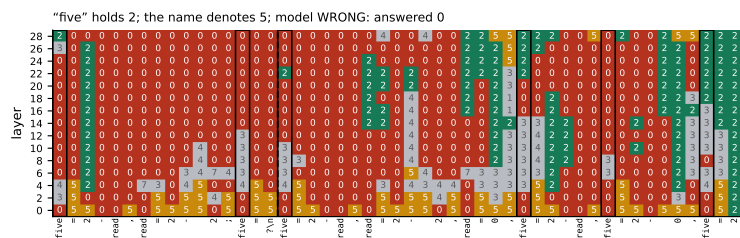
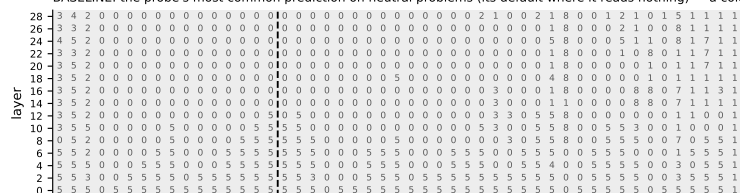
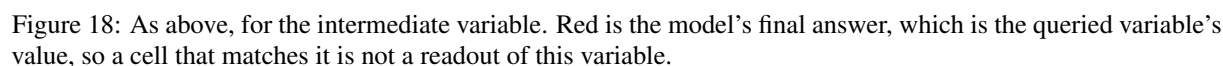
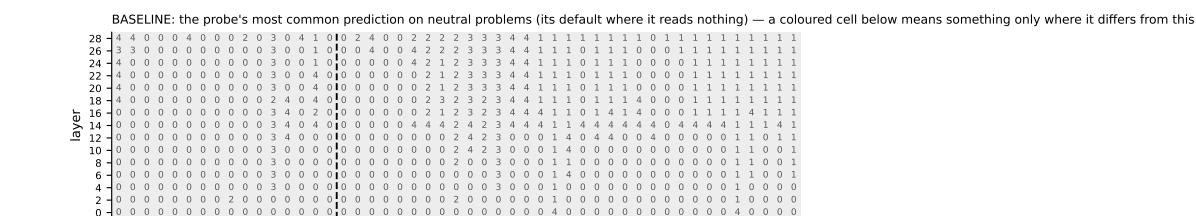


Figure 17: A second such problem.



Top-1 probe prediction at every layer and token (Llama-3.2-3B, L3, with chain of thought; boxed = the name's tokens, dashed = start of the chain)

■ true value
 ■ the model's answer (when wrong)
 ■ what the name denotes
 ■ other

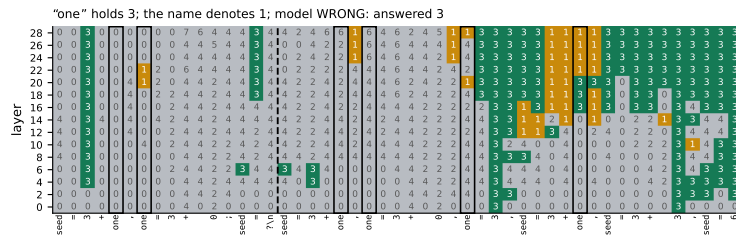
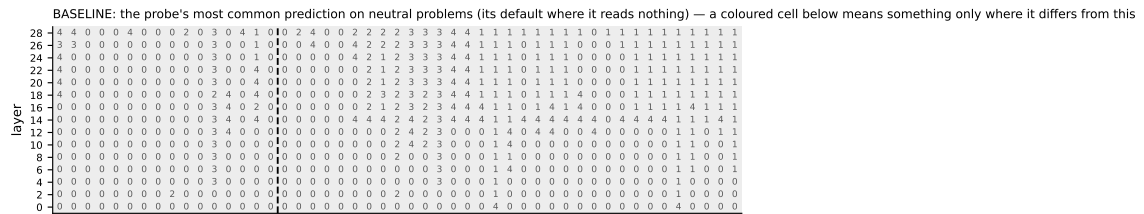


Figure 19: A second problem, intermediate variable.

Llama-3.2-3B, level 3, with chain of thought: per-token probes for the value of the intermediate variable (trained on neutral problems)

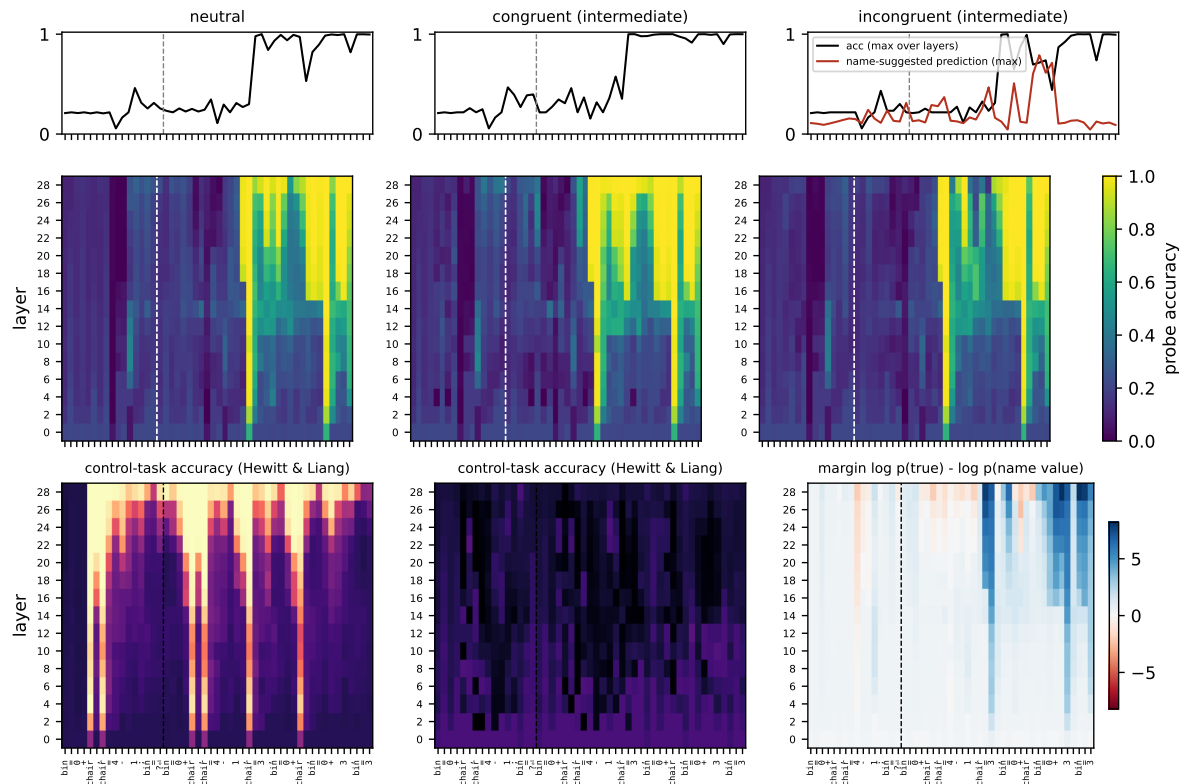


Figure 20: Accuracy heatmaps by token and layer for the neutral, congruent and incongruent conditions, in the layout of Kudo et al.’s Figure 2.

Llama-3.2-3B, level 3, direct answer: per-token probes for the value of the intermediate variable (trained on neutral problems)

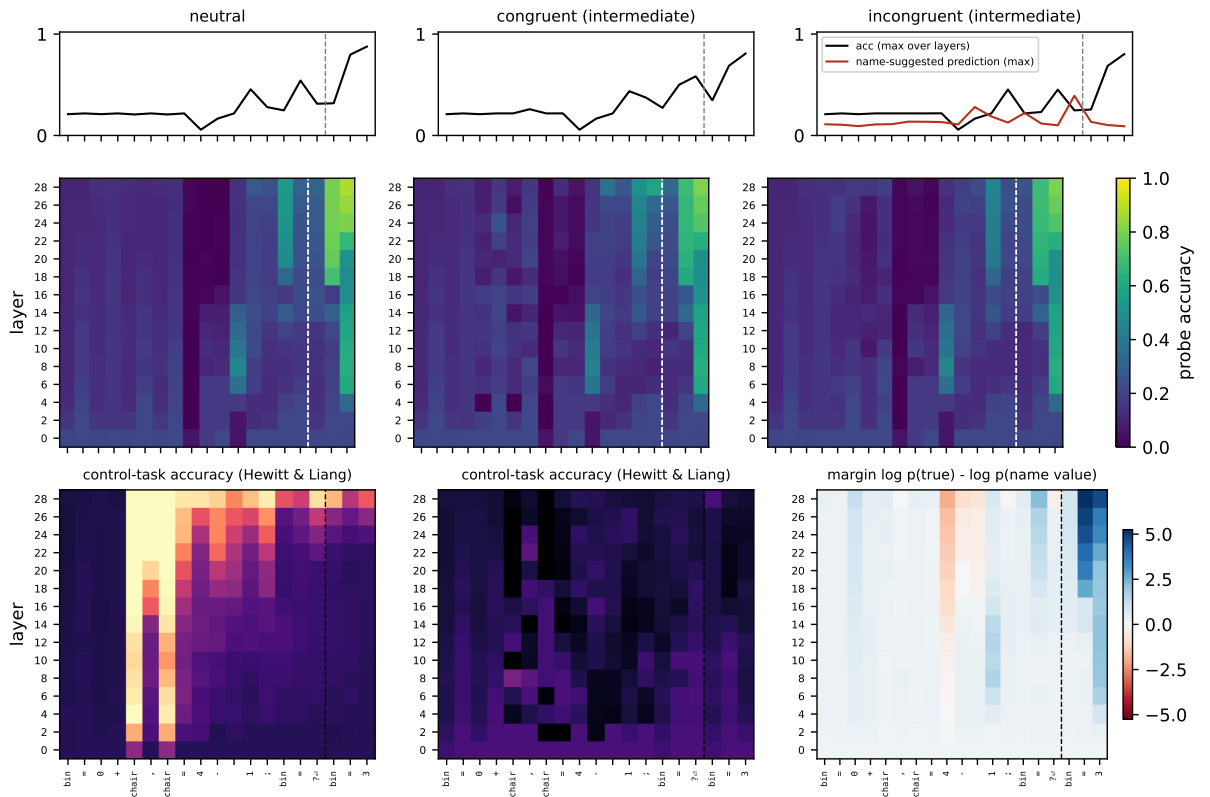


Figure 21: The same heatmaps in the direct regime.

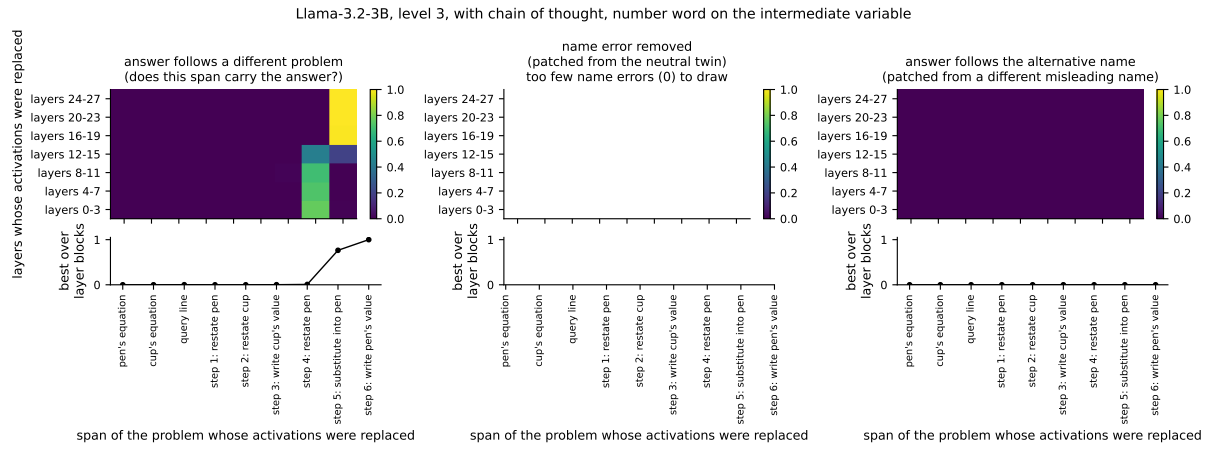


Figure 22: Span-by-layer-window patching grids, three sources (the error-removal panel is empty: with CoT there are too few name errors to draw), after Kudo et al.'s Figures 5–6.

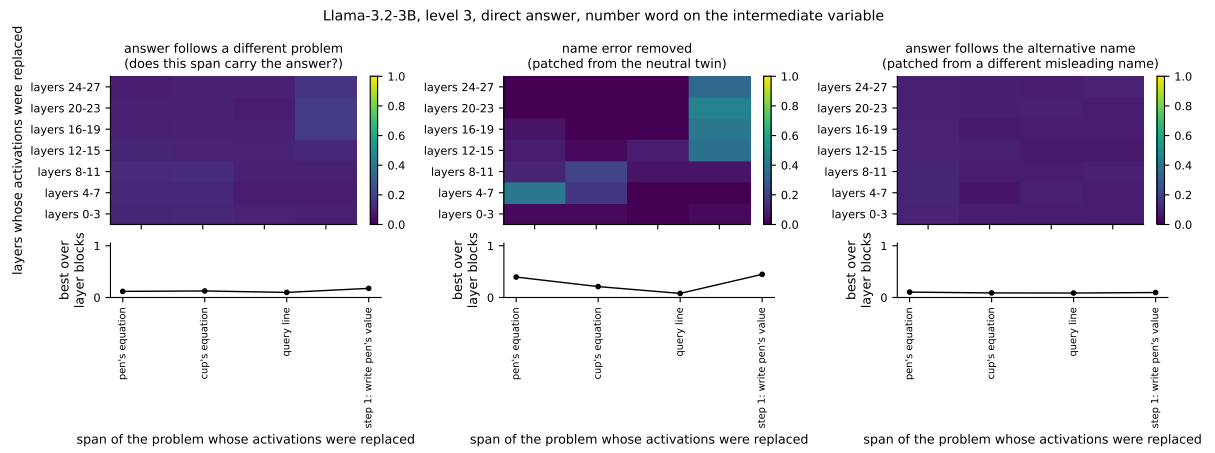


Figure 23: The same grids in the direct regime.

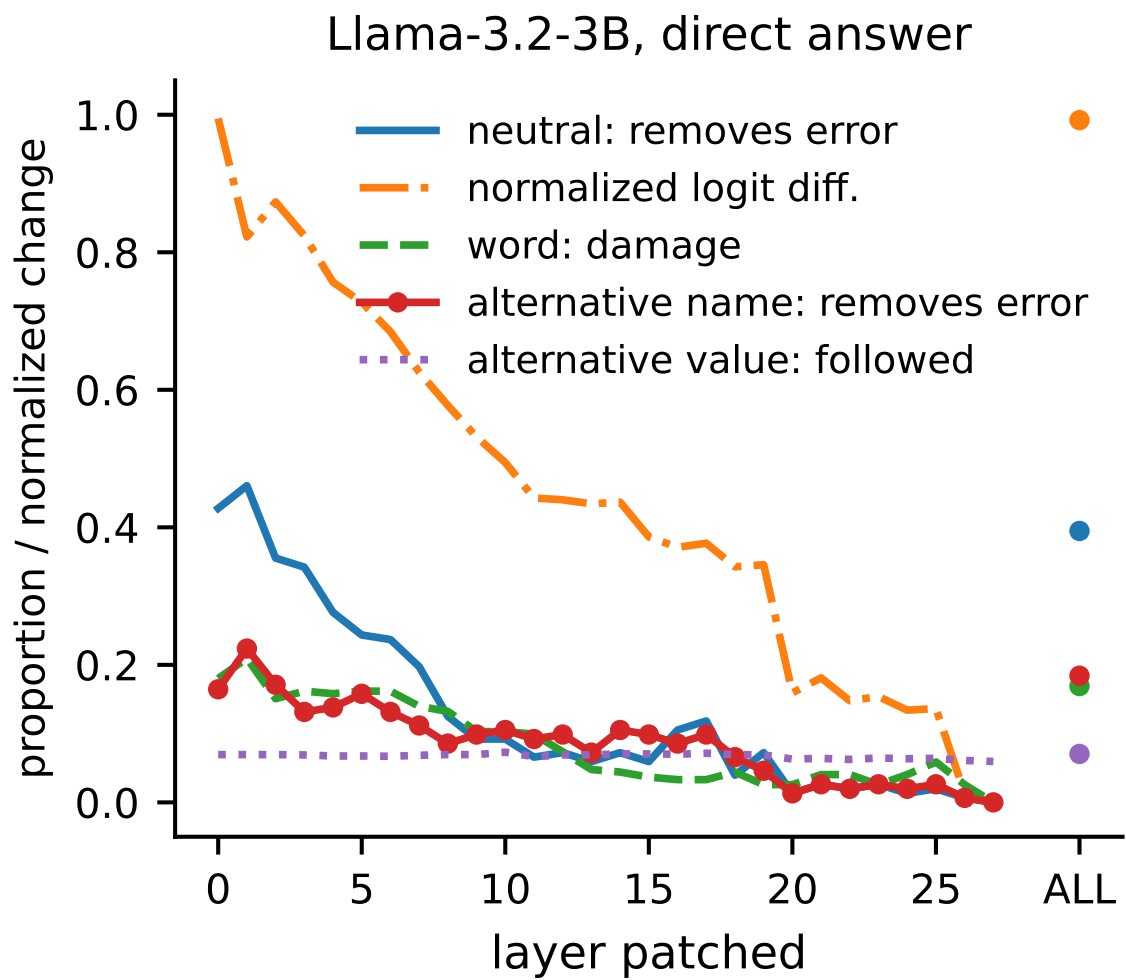


Figure 24: Name errors removed by patching, by layer, with both controls.

Llama-3.2-3B, level 3, with chain of thought: per-token probes for the value of the queried variable (trained on neutral problems)

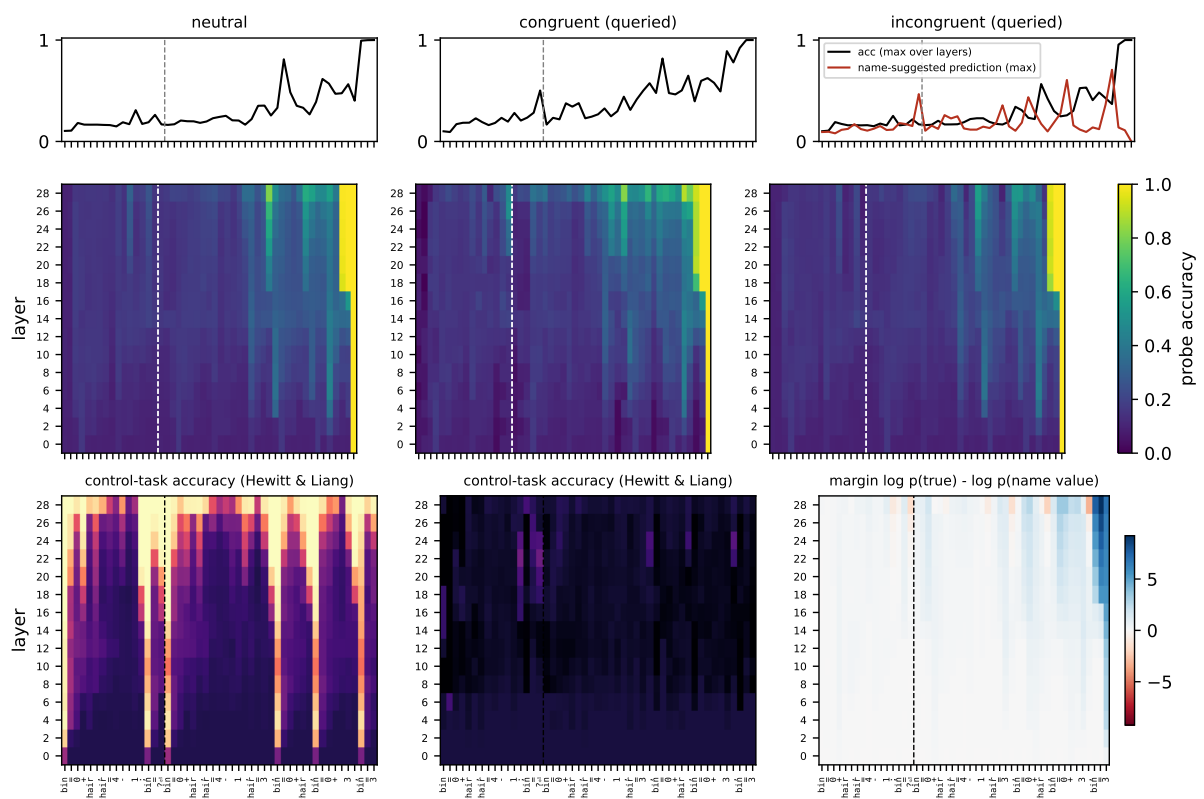


Figure 25: Accuracy heatmaps as above, chain-of-thought regime, queried variable.

Llama-3.2-3B, level 3, direct answer: per-token probes for the value of the queried variable (trained on neutral problems)

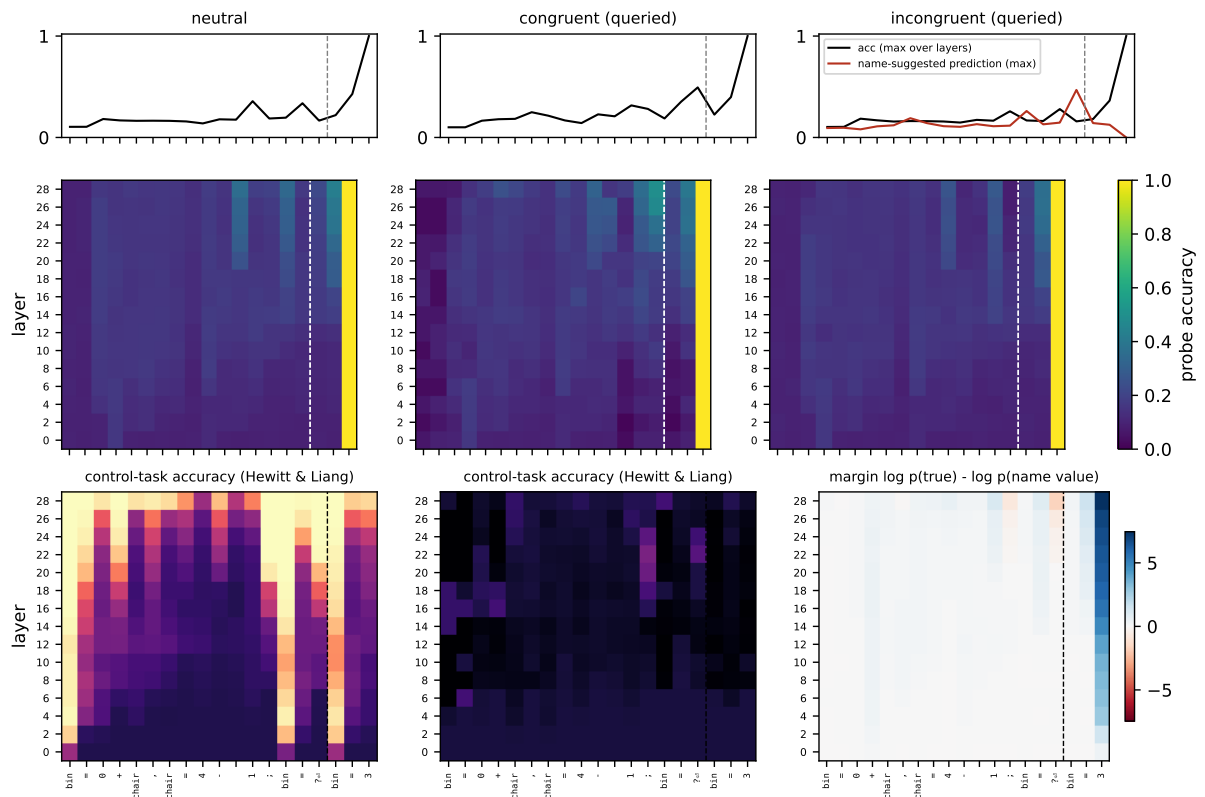


Figure 26: The same heatmaps in the direct regime, queried variable.

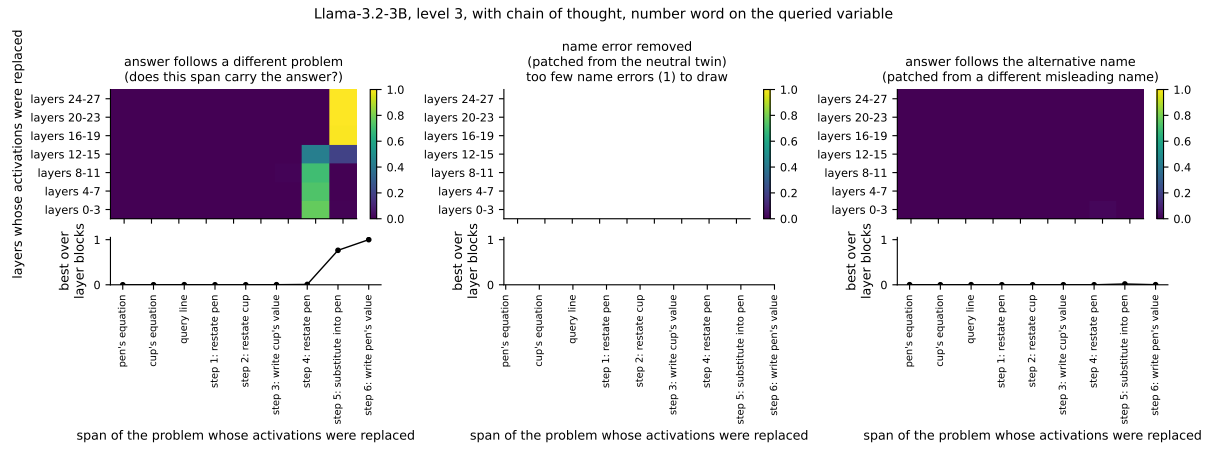


Figure 27: Span-by-layer-window patching grids, chain-of-thought regime, queried variable; the error-removal panel is empty for the same reason.

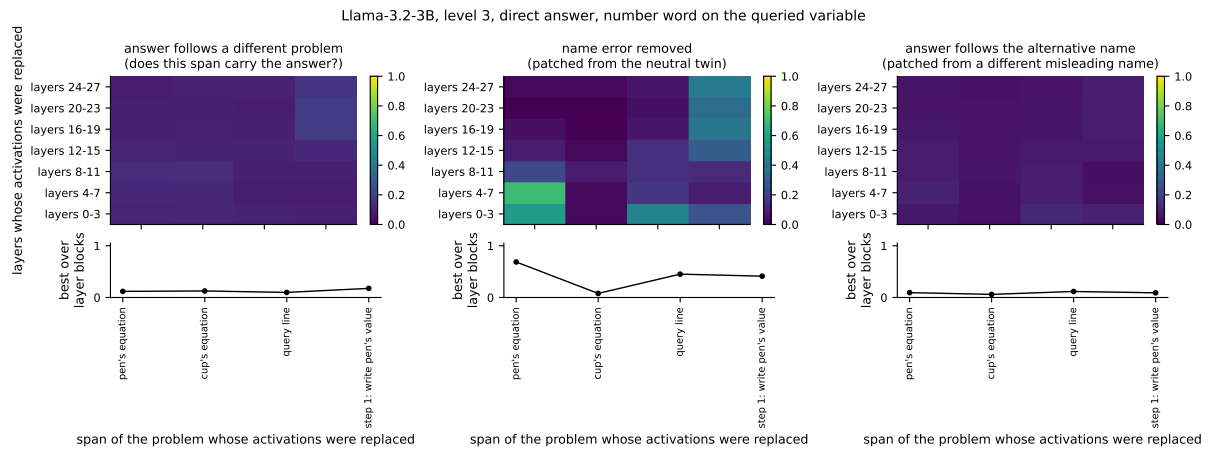


Figure 28: The same grids in the direct regime, queried variable.

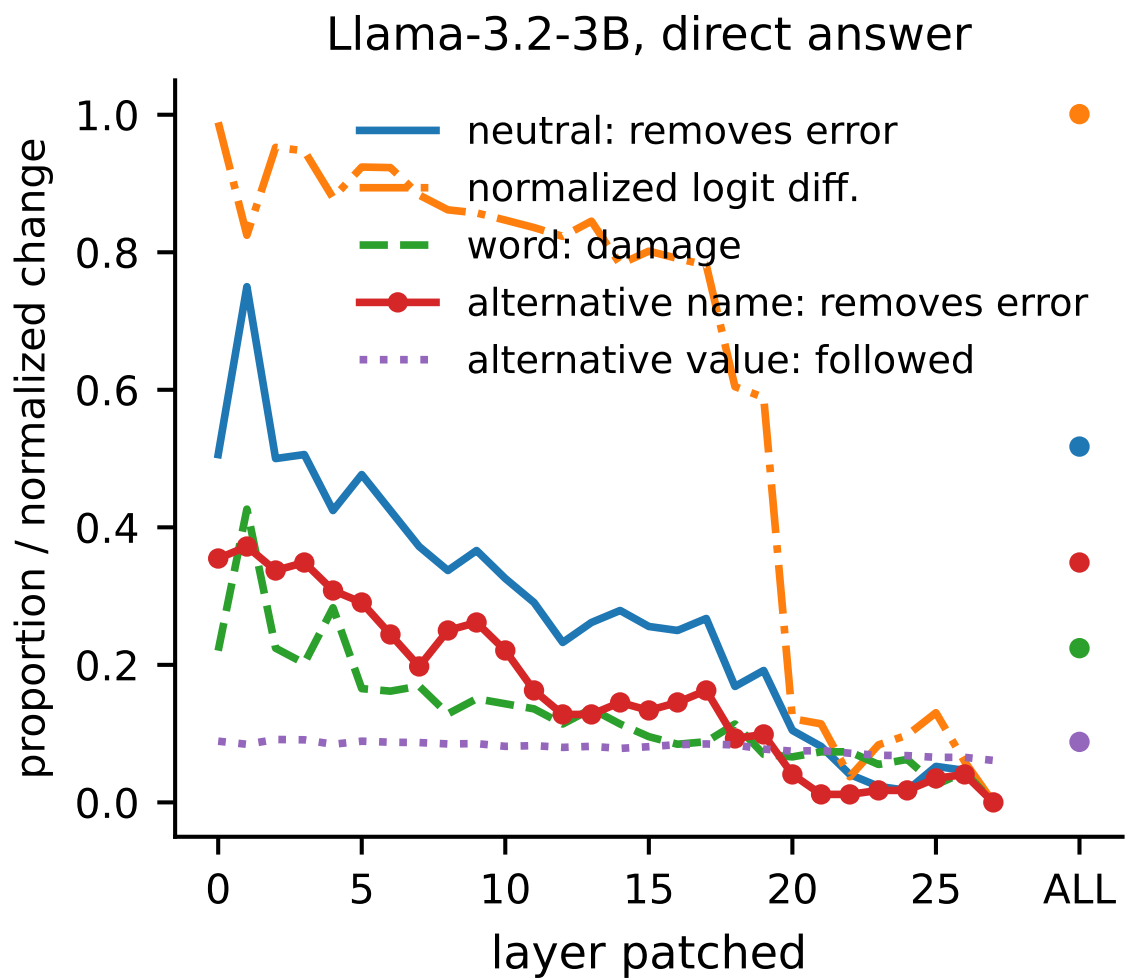


Figure 29: Name errors removed by patching, by layer, with both controls, queried variable.

Top-1 probe prediction over the model's own chain (Llama-3.2-3B, L3, with chain of thought; probes trained at the value step of "v1" on neutral problems; 8 instances)

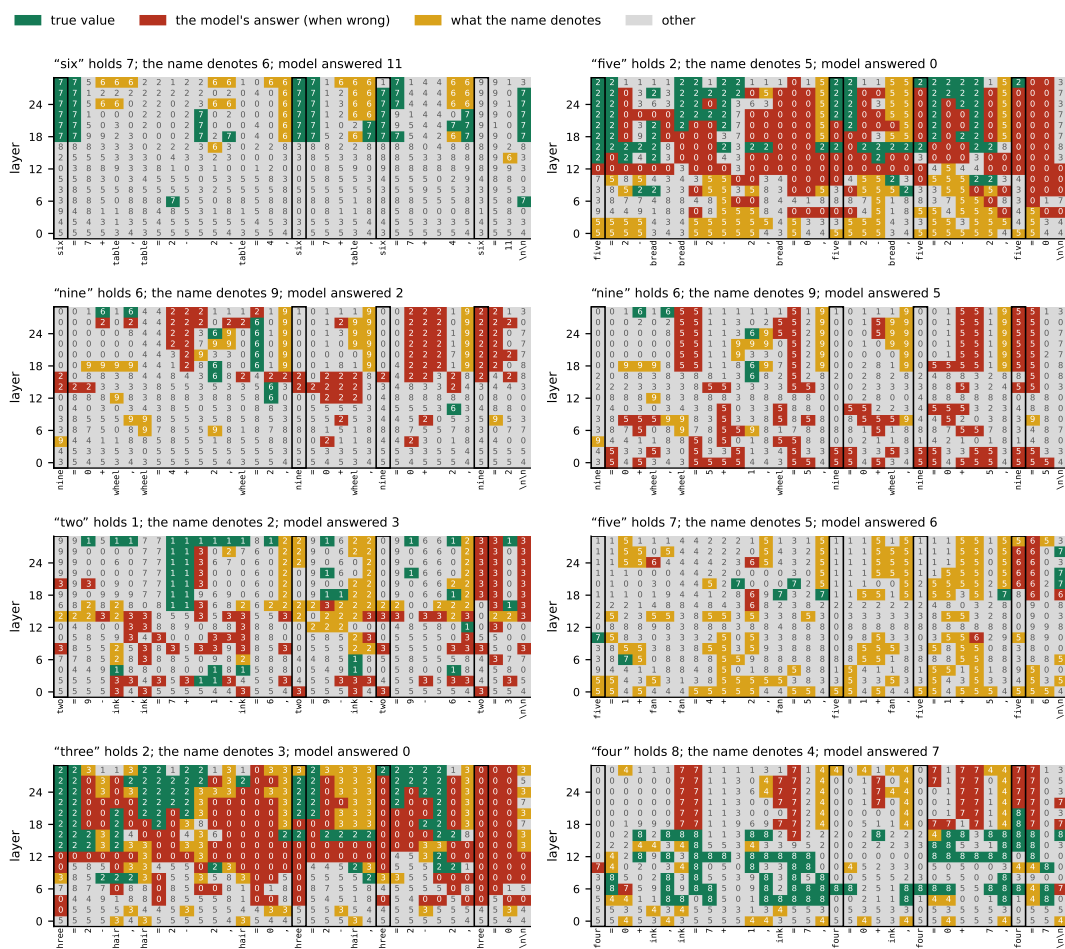


Figure 30: The own-chain probe figure at full size.

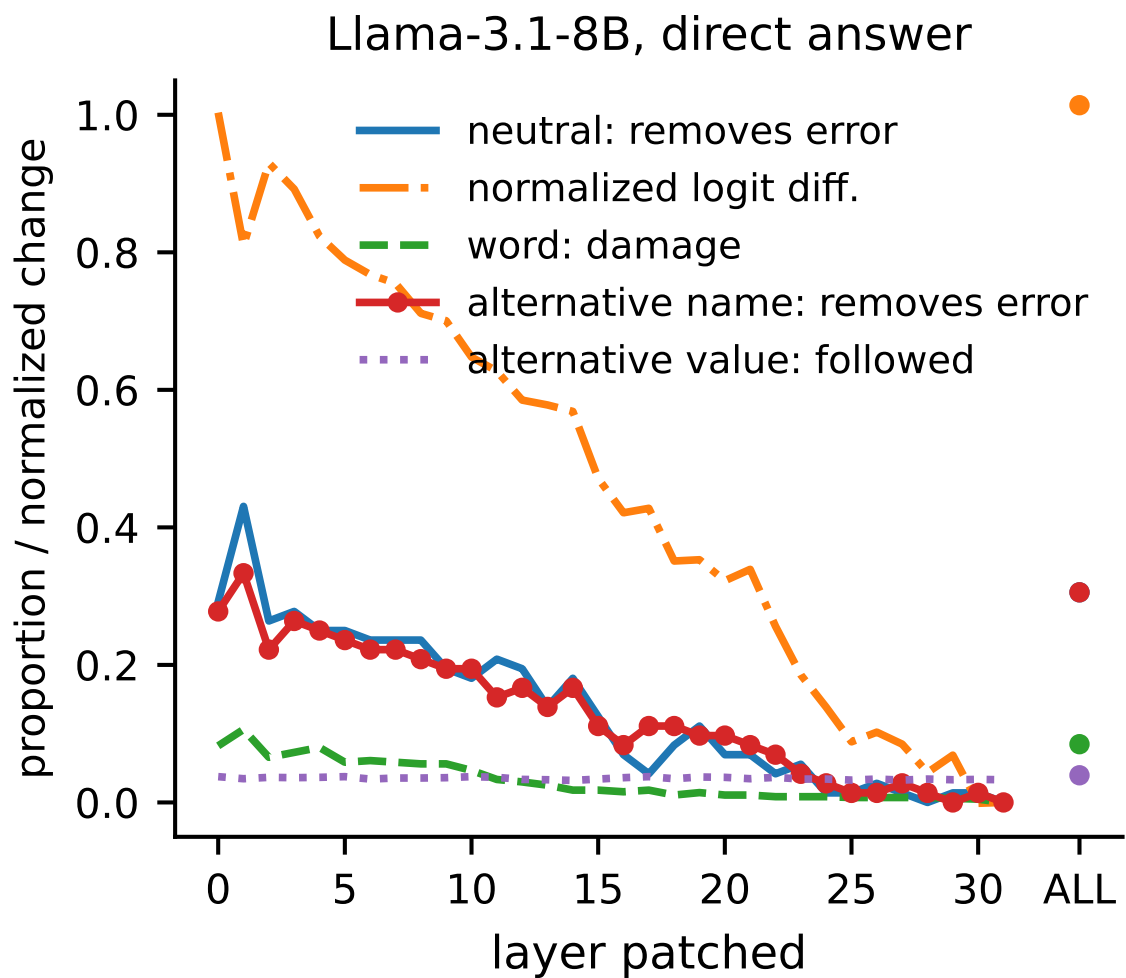


Figure 31: Llama-3.1-8B, name errors removed by patching, by layer, with both controls, direct regime, intermediate variable.

Llama-3.1-8B, level 3, with chain of thought: per-token probes for the value of the queried variable (trained on neutral problems)

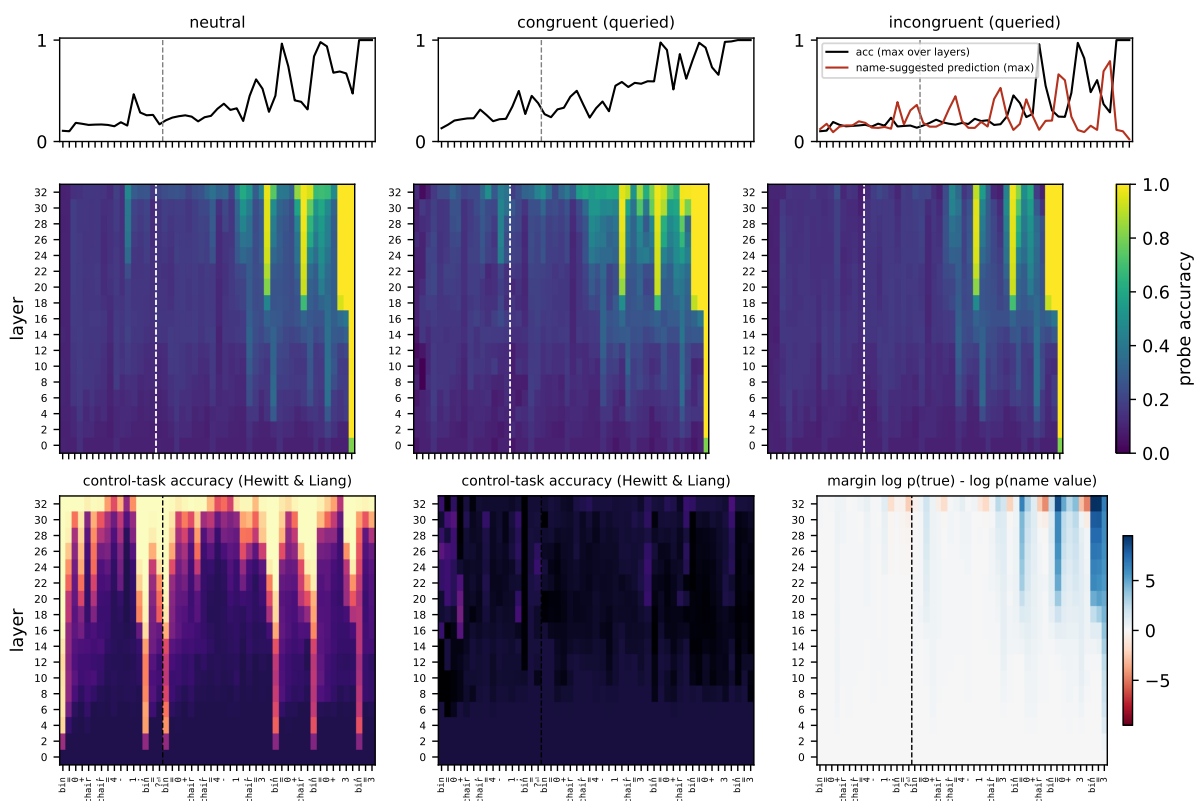


Figure 32: Llama-3.1-8B, chain-of-thought regime, queried variable: the accuracy heatmaps of the Llama-3.2-3B figures. The value is decodable from the equation that restates the variable’s definition, before the chain writes it.

Llama-3.1-8B, level 3, with chain of thought: per-token probes for the value of the intermediate variable (trained on neutral problems)

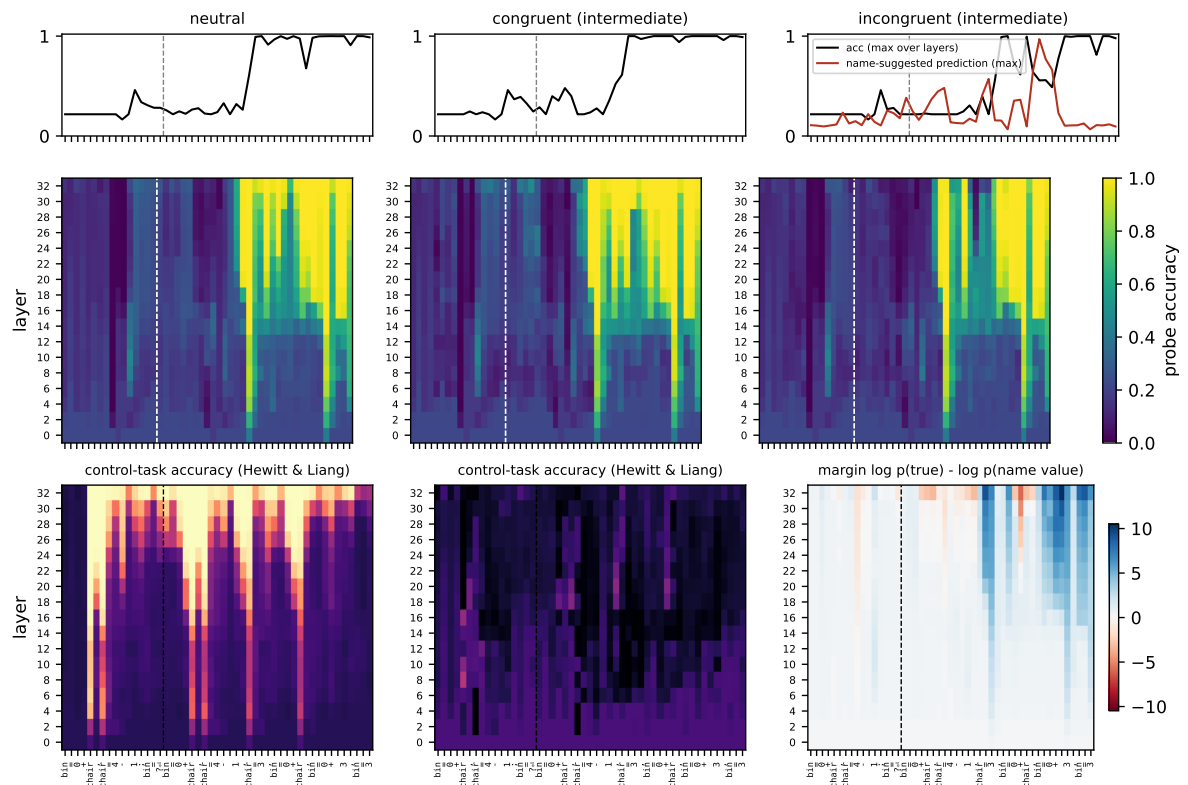


Figure 33: Llama-3.1-8B, chain-of-thought regime, intermediate variable.

Llama-3.1-8B, level 3, direct answer: per-token probes for the value of the intermediate variable (trained on neutral problems)

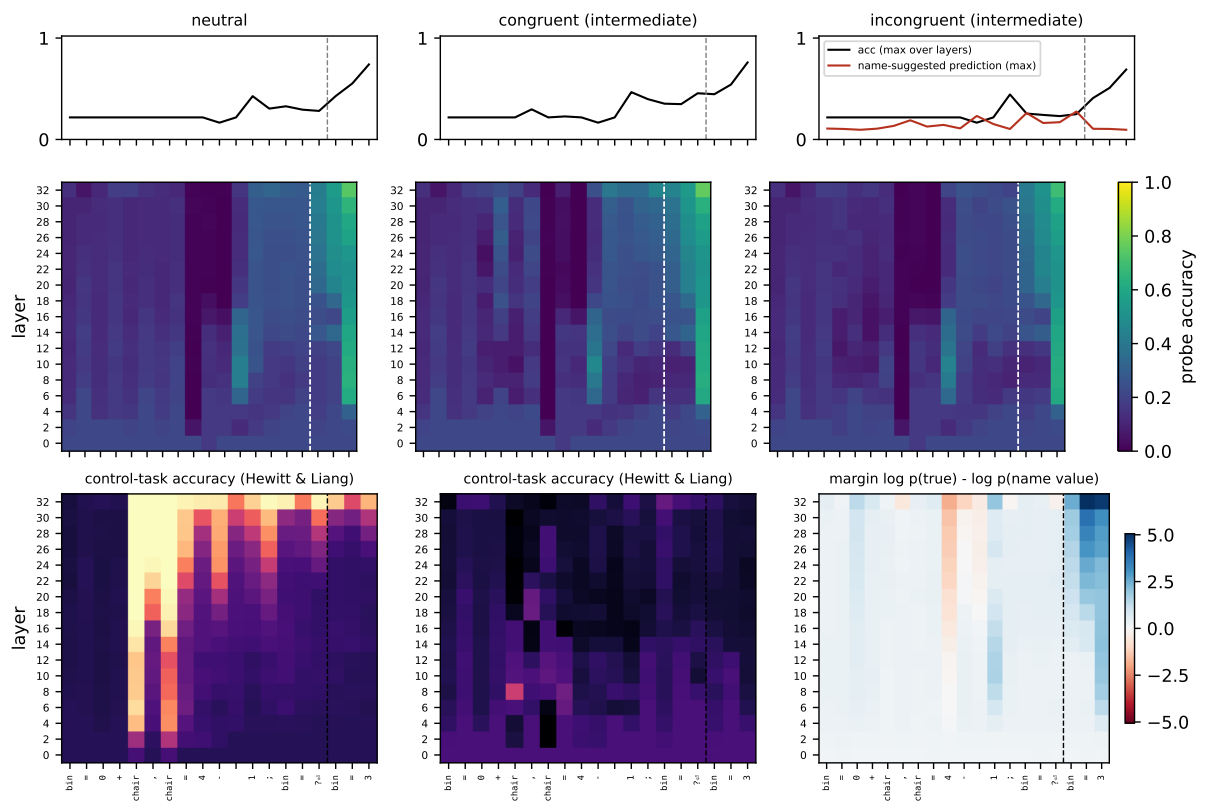


Figure 34: Llama-3.1-8B, direct regime, intermediate variable.

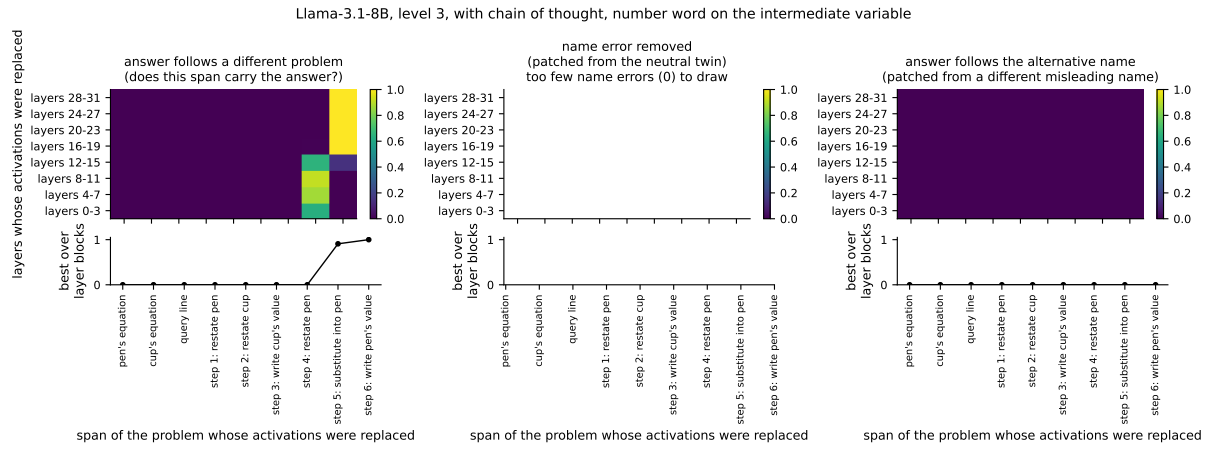


Figure 35: Llama-3.1-8B, span-by-layer-window patching grids, chain-of-thought regime. The error-removal panel is empty: none of the 500 patched problems is answered with the name-suggested value.

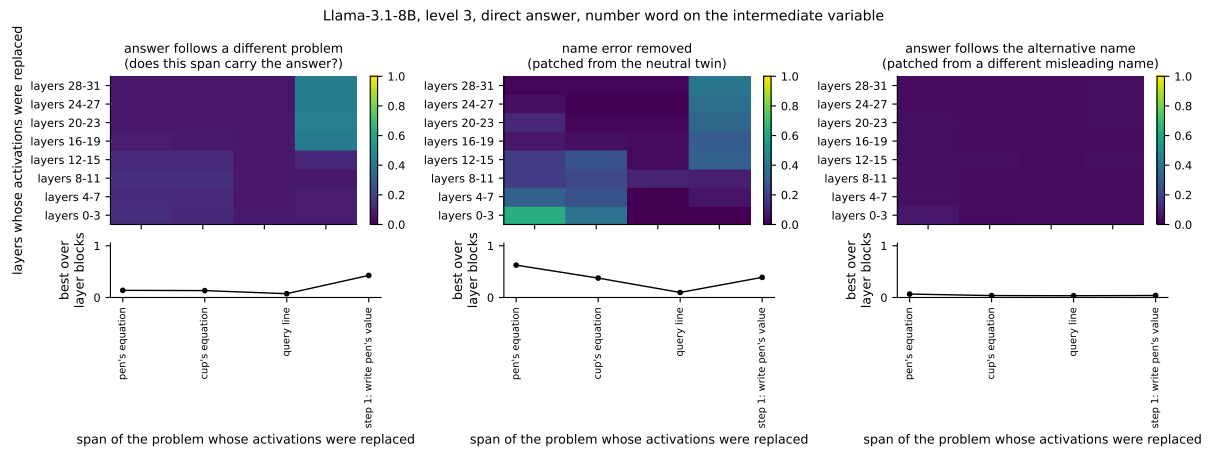


Figure 36: Llama-3.1-8B, the same grids in the direct regime.

Llama-3.1-8B, level 3, direct answer: per-token probes for the value of the queried variable (trained on neutral problems)

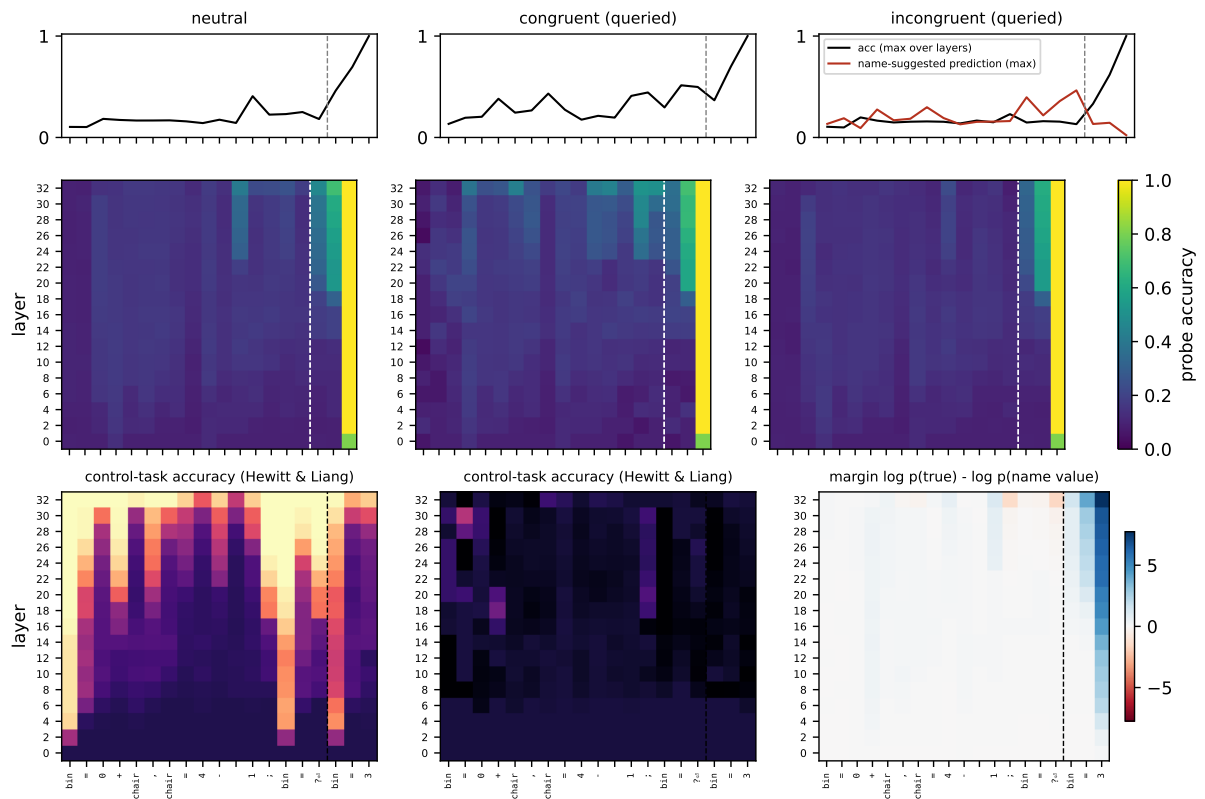


Figure 37: Llama-3.1-8B, direct regime, queried variable.

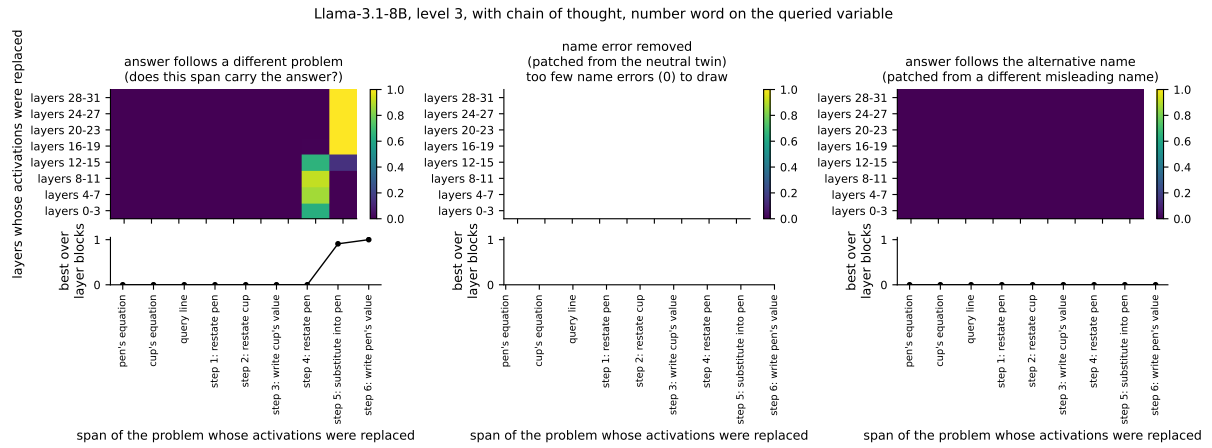


Figure 38: Llama-3.1-8B, patching grids, chain-of-thought regime, queried variable; the error-removal panel is empty for the same reason.

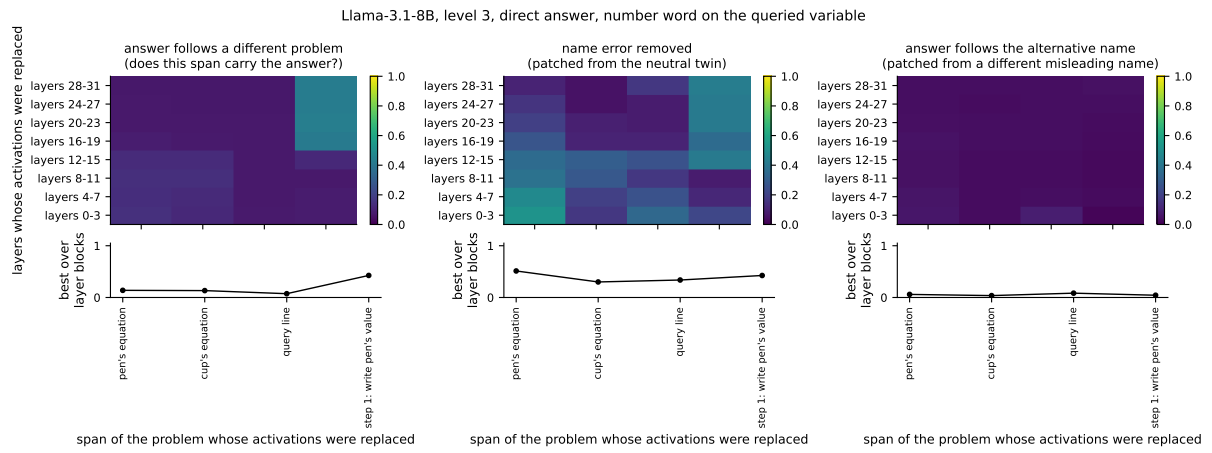


Figure 39: Llama-3.1-8B, patching grids, direct regime, queried variable.

L Task competence and value readouts

Table 11 compares both variables for 9 models. Layers are selected by neutral selectivity, averaging three probe seeds. Task accuracy and paired excess name errors pool demonstration sets 7, 11 and 13; probes use set 7. The within-model difference holds overall task accuracy fixed but does not isolate a causal effect of the readout. We average the two variable readouts within each model before computing descriptive correlations. Table 12 repeats them after omitting each model. Roles are not independent observations. With this small, heterogeneous panel, correlations cannot distinguish overall competence from the role of decodable information.

M Gold-trained transfer to generated chains

We read the state before the first numeric assignment to the target in each saved generated chain. Expression operands are not counted as completed value writes; tokens that include any part of the supplied value are excluded. A fixed sample of 200 matched sets is selected by a hash independent of outcomes, shared across models, variables and demonstration sets. All additional chains with name errors and their neutral twins are evaluated separately. Neutral gold-chain probes use 10,000 separate training problems, three probe seeds and the original neutral-selectivity layer. Table 14 tests transfer to held-out neutral generated chains. A high-accuracy interpretation requires at least 90% neutral calibration in that demonstration set. Cases below that threshold yield descriptive readouts only. The augmentation is selected by observed errors and cannot estimate their population rate. Appendix N separately trains probes on generated neutral chains and tests the same held-out prefixes. Llama-3.2-3B meets the neutral transfer threshold in 6 of 6 cells, with calibration accuracy from 97.2 to 100.0%. OLMo-2-1B meets the neutral transfer threshold in 0 of 6 cells, with calibration accuracy from 19.4 to 62.2%. Name-error cases in the competent model are few; error-conditioned readouts cannot establish a general mechanism from that subset. Failed neutral calibration prevents a strong generated-chain interpretation for the exception model.

N Training probes on generated neutral chains

Gold-trained probe transfer (Appendix M) may fail because the states along generated chains differ from those along the supplied correct chain. We therefore train on freely generated neutral chains from 2,000 source problems and select layers on 400 separate validation problems. Canonical equations partition the source pool, keeping renamed versions of a computation together. The training and validation pools contain 630 and 155 computations; both exclude all test and demonstration computations. These counts distinguish independent computations from repeated name renderings. Generation is greedy with demonstration set 7 and a 128-token limit. The probe sees the hidden state before the target variable’s first numeric assignment, with every token boundary checked to exclude the value digit. Missing or unsafe boundaries are excluded and counted. We retain all usable neutral chains regardless of whether the model writes the correct value, using the executed true value as the training label. Three probe seeds use the original full-batch SGD recipe (learning rate 0.001, 10,000 epochs). States are stored in float32 and nonfinite states or fitted weights are rejected. Layers are searched at stride two, including the final layer, and chosen only by mean validation accuracy with the lowest-layer tie break. The name-identity control is trained separately and does not select layers. Table 15 reports the usable source counts and selected layers. Evaluation uses exactly the earlier fixed 200-set sample and separately reported additional name-error cases under demonstration sets 7, 11 and 13. Table 16 gives held-out neutral calibration and boundary coverage. Intervals use 4,000 canonical-computation bootstrap draws, keeping repeated names together. The 90% calibration threshold applies to the point estimate, not the lower interval endpoint. Llama passes all six calibration cells; OLMo passes none. Retraining on generated chains therefore does not validate a strong readout account of OLMo’s own errors. Table 17 retains its error readouts descriptively. Llama has only one name-error case in the primary sample and thirteen additional observed writes; these rare, selected cases cannot establish a general mechanism. Paired changes from gold-trained transfer are shown in Table 18; they compare the same eligible prefixes, not independent model samples.

Model	Role	Layer	Task	Neutral probe	Misleading probe	Excess
Gemma-3-12B-I	v1	40	100.0	100.0	100.0	0.0
Gemma-3-12B-I	v2	42	100.0	100.0	100.0	0.0
Gemma-3-4B	v1	25	100.0	100.0	100.0	0.0
Gemma-3-4B	v2	28	100.0	99.8	100.0	0.0
Gemma-3-4B-I	v1	29	100.0	100.0	99.9	0.0
Gemma-3-4B-I	v2	28	100.0	99.8	99.9	0.0
Llama-3.1-8B	v1	20	95.2	100.0	100.0	-0.1
Llama-3.1-8B	v2	22	95.2	99.4	99.2	-0.0
Llama-3.1-8B-I	v1	20	99.7	100.0	100.0	0.0
Llama-3.1-8B-I	v2	20	99.7	100.0	100.0	0.0
Llama-3.2-3B	v1	19	98.3	99.9	100.0	0.1
Llama-3.2-3B	v2	21	98.3	97.6	99.6	-0.2
Llama-3.2-3B-I	v1	18	100.0	99.9	100.0	0.0
Llama-3.2-3B-I	v2	19	100.0	99.9	99.9	0.0
OLMo-2-1B	v1	14	47.2	99.7	100.0	0.4
OLMo-2-1B	v2	0	47.2	21.8	21.8	3.2
OLMo-2-7B	v1	22	95.3	100.0	100.0	-0.1
OLMo-2-7B	v2	26	95.3	99.4	97.6	-0.0

Table 11: Level-3 arithmetic task and probe accuracies (%) and paired excess name errors (points). Neutral task accuracy is shared by the two roles. All available complete model pairs are shown; below-floor models remain descriptive.

Omitted model	Task-excess	Probe-excess	Task-probe
All models	-0.99	-1.00	+1.00
Gemma-3-12B-I	-0.99	-1.00	+1.00
Gemma-3-4B	-0.99	-1.00	+1.00
Gemma-3-4B-I	-0.99	-1.00	+1.00
Llama-3.1-8B	-0.99	-1.00	+1.00
Llama-3.1-8B-I	-0.99	-1.00	+1.00
Llama-3.2-3B	-0.99	-1.00	+1.00
Llama-3.2-3B-I	-0.99	-1.00	+1.00
OLMo-2-1B	+0.98	+0.78	+0.85
OLMo-2-7B	-0.99	-1.00	+1.00

Table 12: Descriptive Pearson correlations of model means; Spearman values and within-model variable differences are in the saved summary. No significance or causal claim is based on these coefficients.

Model	Neutral probe	Misleading probe	Excess
Gemma-3-12B-I	+0.0	+0.0	+0.0
Gemma-3-4B	-0.2	+0.0	+0.0
Gemma-3-4B-I	-0.2	+0.0	+0.0
Llama-3.1-8B	-0.6	-0.8	+0.1
Llama-3.1-8B-I	+0.0	+0.0	+0.0
Llama-3.2-3B	-2.2	-0.4	-0.3
Llama-3.2-3B-I	+0.1	-0.1	+0.0
OLMo-2-1B	-77.9	-78.2	+2.7
OLMo-2-7B	-0.5	-2.3	+0.1

Table 13: Intermediate minus queried variable, in percentage points. Overall neutral task accuracy is identical within each model. These differences are descriptive comparisons between variable positions.

Model	Set	Role	Neutral coverage	Calibration	Sample error probe	Extra errors n	Extra error probe
OLMo-2-1B	7	v1	194/200	62.2 [55.2, 68.7]	0.0 [0.0, 0.0]	69	1.4 [0.0, 3.9]
OLMo-2-1B	7	v2	186/200	19.4 [14.0, 25.3]	54.5 [27.3, 81.8]	82	65.9 [56.1, 75.6]
OLMo-2-1B	11	v1	187/200	55.3 [48.3, 62.6]	0.0 [0.0, 0.0]	59	5.6 [0.6, 12.4]
OLMo-2-1B	11	v2	129/200	21.7 [14.7, 28.7]	100.0 [100.0, 100.0]	21	66.7 [47.6, 85.7]
OLMo-2-1B	13	v1	193/200	52.7 [45.6, 59.8]	27.5 [7.8, 49.0]	159	10.7 [6.5, 15.3]
OLMo-2-1B	13	v2	124/200	24.2 [16.9, 32.3]	100.0 [100.0, 100.0]	30	60.0 [43.3, 76.7]
Llama-3.2-3B	7	v1	200/200	98.0 [96.0, 99.5]	0.0 [0.0, 0.0]	8	12.5 [0.0, 37.5]
Llama-3.2-3B	7	v2	200/200	97.2 [94.8, 99.0]	–	0	–
Llama-3.2-3B	11	v1	199/200	100.0 [100.0, 100.0]	–	0	–
Llama-3.2-3B	11	v2	200/200	100.0 [100.0, 100.0]	–	0	–
Llama-3.2-3B	13	v1	199/200	99.5 [98.5, 100.0]	–	5	0.0 [0.0, 0.0]
Llama-3.2-3B	13	v2	200/200	99.5 [98.5, 100.0]	–	0	–

Table 14: Generated-chain true-digit accuracy (%) [95% interval]. Sample error probe conditions on name errors in the fixed sample; extra name-error cases exclude that sample. Dashes indicate no eligible cases. Saved summaries record all missing or excluded boundaries and calibration decisions.

Model	Role	Train n	Validation n	Layer	Validation (%)	Name ID (%)
OLMo-2-1B-I	v1	1924	390	14	47.9	41.6
OLMo-2-1B-I	v2	1877	376	16	87.8	86.6
Llama-3.2-3B	v1	1994	398	22	97.5	26.8
Llama-3.2-3B	v2	2000	400	20	99.2	28.6

Table 15: Probe source rows with usable generated boundaries, selected layers and validation scores. Source cohorts are computation-disjoint; usable counts vary by model and variable. Validation scores select layers and are not test estimates.

Model	Role	Set	Neutral coverage	Calibration (%)	Name ID (%)	Pass
OLMo-2-1B-I	v1	7	194/200	61.3 [54.1, 68.6]	37.3 [31.2, 43.4]	no
OLMo-2-1B-I	v1	11	187/200	52.8 [45.1, 60.3]	32.4 [26.6, 38.5]	no
OLMo-2-1B-I	v1	13	193/200	50.3 [43.0, 57.4]	35.2 [29.1, 41.6]	no
OLMo-2-1B-I	v2	7	186/200	78.0 [71.5, 83.9]	81.9 [76.9, 86.7]	no
OLMo-2-1B-I	v2	11	129/200	80.1 [72.8, 86.8]	56.6 [49.6, 63.3]	no
OLMo-2-1B-I	v2	13	124/200	71.8 [64.2, 78.4]	46.8 [37.8, 55.6]	no
Llama-3.2-3B	v1	7	200/200	98.5 [96.7, 100.0]	18.5 [13.8, 23.7]	yes
Llama-3.2-3B	v1	11	199/200	100.0 [100.0, 100.0]	13.9 [9.5, 18.5]	yes
Llama-3.2-3B	v1	13	199/200	97.2 [94.2, 99.5]	16.9 [12.3, 21.8]	yes
Llama-3.2-3B	v2	7	200/200	96.3 [93.6, 98.6]	28.8 [22.7, 35.0]	yes
Llama-3.2-3B	v2	11	200/200	100.0 [100.0, 100.0]	17.2 [12.4, 22.1]	yes
Llama-3.2-3B	v2	13	200/200	97.7 [95.5, 99.3]	17.8 [13.3, 22.8]	yes

Table 16: Held-out primary neutral generated chains. Coverage counts valid pre-value boundaries over all primary neutral rows. Accuracy intervals are 95% canonical-computation bootstrap intervals; pass means the calibration point estimate meets 90%.

Model	Role	Set	Sample n	Sample accuracy	Extra n	Extra accuracy
OLMo-2-1B-I	v1	7	8	4.2 [0.0, 12.5]	69	2.9 [0.0, 7.0]
OLMo-2-1B-I	v1	11	7	19.0 [0.0, 38.1]	59	10.7 [4.2, 18.8]
OLMo-2-1B-I	v1	13	17	21.6 [7.8, 39.2]	159	13.6 [8.6, 19.4]
OLMo-2-1B-I	v2	7	11	33.3 [15.2, 51.5]	82	18.7 [11.4, 27.2]
OLMo-2-1B-I	v2	11	2	0.0 [0.0, 0.0]	21	17.5 [3.3, 31.8]
OLMo-2-1B-I	v2	13	1	0.0 [0.0, 0.0]	30	25.6 [13.7, 39.1]
Llama-3.2-3B	v1	7	1	0.0 [0.0, 0.0]	8	12.5 [0.0, 42.9]
Llama-3.2-3B	v1	11	0	–	0	–
Llama-3.2-3B	v1	13	0	–	5	0.0 [0.0, 0.0]
Llama-3.2-3B	v2	7	0	–	0	–
Llama-3.2-3B	v2	11	0	–	0	–
Llama-3.2-3B	v2	13	0	–	0	–

Table 17: True-digit accuracy (%) [95% interval] on actual name errors, with the fixed sample and additional error-selected cases separated. Extra cases exclude the primary sample. Dashes denote empty strata. OLMo fails neutral calibration, so its readouts are descriptive.

Model	Role	Set	Neutral change	Sample error change	Extra error change
OLMo-2-1B-I	v1	7	-0.9 [-4.9, 3.4]	4.2 [0.0, 12.5]	1.4 [-1.0, 5.3]
OLMo-2-1B-I	v1	11	-2.5 [-7.7, 2.7]	19.0 [0.0, 38.1]	5.1 [-0.6, 12.1]
OLMo-2-1B-I	v1	13	-2.4 [-7.2, 2.2]	-5.9 [-19.6, 5.9]	2.9 [-0.8, 6.7]
OLMo-2-1B-I	v2	7	58.6 [49.5, 67.5]	-21.2 [-57.6, 12.2]	-47.2 [-63.3, -30.1]
OLMo-2-1B-I	v2	11	58.4 [47.9, 67.9]	-100.0 [-100.0, -100.0]	-49.2 [-71.4, -20.8]
OLMo-2-1B-I	v2	13	47.6 [37.7, 57.1]	-100.0 [-100.0, -100.0]	-34.4 [-55.6, -13.3]
Llama-3.2-3B	v1	7	0.5 [0.0, 1.6]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Llama-3.2-3B	v1	11	0.0 [0.0, 0.0]	—	—
Llama-3.2-3B	v1	13	-2.3 [-5.1, -0.3]	—	0.0 [0.0, 0.0]
Llama-3.2-3B	v2	7	-0.8 [-2.9, 0.9]	—	—
Llama-3.2-3B	v2	11	0.0 [0.0, 0.0]	—	—
Llama-3.2-3B	v2	13	-1.8 [-3.7, -0.5]	—	—

Table 18: Paired accuracy changes in percentage points [95% interval] from gold-trained transfer to generated-neutral training on the same valid prefixes. Error-selection and failed neutral calibration limit the interpretation.